HuntAssist Stage 1 Software and Agent Design

Bug Bounty Traffic Analysis and Investigation Planning Agent

EECS 3311 Software Design
Course Project Stage 1
Fall 2026

Design before implementation

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Executive Summary

HuntAssist is a local-first Java application for an individual bug bounty hunter working on assets for which they have explicit authorization. It imports HTTP traffic captured in Burp Suite, builds a structured attack-surface model, retrieves relevant techniques from the hunter's own methodology notes, and uses a bounded LLM agent to propose evidence-grounded investigations. The hunter reviews and performs every test. The course release does not send live requests, scan targets, or claim that a vulnerability has been confirmed.

The design divides responsibility deliberately. Deterministic Java components parse traffic, enforce program scope, normalize endpoints, redact secrets, compare controlled accounts, retrieve knowledge, preserve evidence, and track coverage. The LLM interprets a goal, selects read-only tools, connects observations to relevant methodology, and explains possible lines of investigation. A validator rejects unsupported references, out-of-scope material, unavailable actions, and statements that present a hypothesis as a confirmed bug.

The application is a modular monolith implemented with Java 21, Spring Boot, two supported local interfaces, H2 file storage, a separate evidence vault, and a Lucene knowledge index. Spring MVC and Thymeleaf provide the browser GUI; Picocli provides the command-line interface. Both are inbound adapters over the same application services. Ports isolate Burp file formats, Notion or local notes, the LLM provider, storage, and future integrations.

HuntAssist is a multi-vulnerability investigation platform. Vulnerability-specific analysis enters through one compile-time, Spring-managed VulnerabilityModule contract. The course release implements XSS as the first reference module because the supplied XSS methodology provides a detailed, context-sensitive vertical slice; XSS validates the shared contract but does not define the product boundary.

The supported CLI exposes the agent's major workflows for interactive use and automation with human-readable or versioned JSON output. It cannot bypass the safeguards used by the GUI. The course release allows one process at a time to own a project's H2 database and evidence vault, so huntassist serve and standalone CLI commands use an exclusive project-store lock.

The first release focuses on imported Burp traffic and exported notes. Direct Notion synchronization and a Burp Montoya extension are planned adapters. The design therefore serves the course deliverable now and can continue growing into a practical hunting tool afterward.

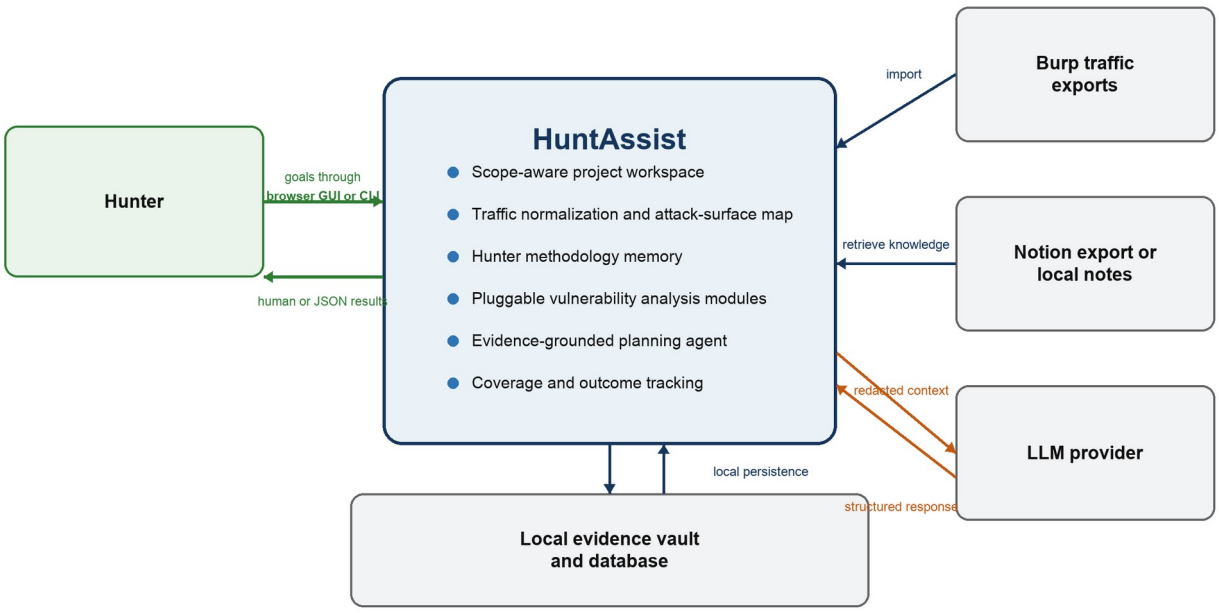


Figure 1 HuntAssist system context and responsibility boundary

1 Project Description and Scope

1.1 Problem

A bug bounty hunter collects a large amount of fragmented information while exploring a web application. Burp history contains requests and responses, personal notes contain techniques and payload constraints, and the hunter separately remembers which paths and roles have already been tested. On a large application, endpoints, parameters, multi-step workflows, and account states are easy to miss. Generic LLM conversations add another fragment: they can discuss a request, but they do not automatically maintain a verified attack-surface model, program scope, evidence provenance, or durable coverage state.

HuntAssist brings these sources into one project. It treats captured traffic as evidence, personal notes as versioned methodology, and LLM output as a hypothesis that must pass deterministic validation. The application helps the hunter decide where to spend attention while preserving the distinction between an observed fact, an agent interpretation, and a manually verified finding.

1.2 Intended User

The primary user is an individual bug bounty hunter performing authorized testing through a coordinated disclosure or bounty program. The user understands HTTP and Burp Suite, controls the accounts represented in the imported traffic, and remains responsible for program rules and live testing. The course evaluator is a secondary reader of the design, but not an application actor.

1.3 Why an Agent Is Appropriate

The application contains many deterministic tasks, but choosing a useful investigation path requires interpretation. The same parameter may deserve different tests depending on where it appears, how the response handles it, which role made the request, what the hunter has already tried, and which methodology sections apply. An LLM is useful for connecting that context and explaining a plan. It is not trusted to parse files, decide scope, handle secrets, establish evidence, or confirm a vulnerability. Those responsibilities remain in Java services.

1.4 Scope of the Course Release

- Create a local project for one bounty program and define authorized asset rules and restrictions.
- Import Burp XML and HAR traffic, with raw HTTP file support as time permits.
- Normalize and deduplicate requests while preserving the original evidence record.
- Build an endpoint, parameter, content-type, GraphQL, authentication-state, and account-role inventory.
- Import Markdown, HTML, text, or a Notion export as methodology knowledge.
- Retrieve relevant methodology sections for an endpoint, input context, category, or question.
- Select an installed vulnerability module and convert scoped evidence into category-specific deterministic observations; the course release supplies the XSS reference module.
- Compare traffic from two controlled test accounts or roles.
- Generate ranked, evidence-linked investigation hypotheses through read-only agent tools.
- Track coverage, hunter decisions, outcomes, and source versions across sessions.
- Access major workflows through either the local browser GUI or a supported Java CLI, with both entry paths enforcing the same scope, privacy, validation, persistence, and audit rules.

1.5 Deferred and Excluded Work

Direct Notion synchronization, real-time Burp ingestion through Montoya, controlled request replay, local-model support, project sharing, additional capture formats, and additional vulnerability modules are planned extensions behind existing interfaces. Candidate modules include access control and IDOR, SQL injection, SSRF, authentication and session handling, CSRF, file upload, GraphQL, business logic, and race conditions. They are not promised for the first implementation.

The course release excludes autonomous exploitation, live scanning, brute force, credential attacks, denial of service, destructive payloads, testing outside the active program scope, automatic vulnerability confirmation, and automatic report submission. It also does not promise to reconstruct encrypted HTTPS from an ordinary PCAP without session secrets.

1.6 Assumptions and Constraints

- The hunter imports only material obtained during authorized testing and defines the active program scope accurately.
- The first version is a single-user application running locally on the hunter's computer.
- Java 21 is used for compatibility with the planned Spring application and a future Burp Montoya adapter.
- huntassist serve launches the browser application; other subcommands boot the same Spring application core in non-web mode.
- The course release allows one process to own a project store at a time. Simultaneous GUI and standalone CLI access is deferred behind a future loopback client mode.
- Provider credentials and other secrets come from local configuration or a secure credential source and are never accepted as command-line arguments.
- Imported notes are reference data, not trusted instructions for the LLM.
- Vulnerability modules analyze already imported, scoped evidence only and cannot send requests to a target, access repositories directly, or call an LLM.
- Recommendations remain unverified hypotheses until the hunter performs and evaluates a test.
- A remote LLM receives minimized and redacted context only when the active RemoteModelPolicy explicitly permits that provider; otherwise model-backed features are unavailable and deterministic views continue to work.
- Performance values in this report are design targets and will be measured during implementation.

2 Requirements and Major Features

2.1 Functional Requirements

ID	Requirement
FR-01	Create a hunt project containing the program name, authorized assets, exclusions, and testing restrictions.
FR-02	Classify every imported exchange as in scope, out of scope, or unresolved before it becomes available to the agent.
FR-03	Detect a traffic file format from its contents and parse each supported format into one common HTTP exchange model.
FR-04	Normalize and deduplicate exchanges while preserving links to every original capture record.
FR-05	Build an attack-surface inventory of hosts, paths, methods, parameters, content types, status codes, GraphQL operations, and observed authentication states.
FR-06	Allow the hunter to label traffic by controlled account, role, workflow, and investigation notes.
FR-07	Import methodology documents from Markdown, HTML, plain text, or a Notion export and preserve document and section provenance.
FR-08	Retrieve methodology sections relevant to the selected vulnerability module, endpoint, observed context, or natural-language question.
FR-09	Compare matched observations from two controlled accounts or roles and report factual request and response differences.
FR-10	Generate ranked investigation hypotheses from scoped traffic, retrieved methodology, previous coverage, and the hunter's stated goal.
FR-11	Attach resolvable traffic evidence and methodology citations to every endpoint-specific recommendation.
FR-12	Reject agent output that is malformed, out of scope, unsupported by its cited records, or presented as a confirmed vulnerability.
FR-13	Track coverage by endpoint, vulnerability category, account context, and result state.
FR-14	Record whether the hunter accepted, modified, dismissed, attempted, or completed each recommendation.
FR-15	Answer natural-language questions by using read-only application tools over imported project data.
FR-16	Mask sensitive request and response headers and values before any context is sent to an external LLM.

ID	Requirement
FR-17	Preserve import history, knowledge-source versions, agent runs, tool traces, and hunter decisions.
FR-18	Recover predictably from unsupported files, malformed records, unavailable providers, invalid model output, and storage errors.
FR-19	Enforce a per-project RemoteModelPolicy and block external model calls unless the program policy explicitly permits them.
FR-20	Register and invoke vulnerability-analysis modules through one common interface, with XSS supplied as the first course-release reference module.
FR-21	Provide a supported command-line interface for project setup, traffic and knowledge import, attack-surface queries, module discovery and analysis, investigation planning, natural-language questions, and coverage operations. CLI commands shall invoke the same application services and safeguards as the GUI and support human-readable and JSON output.

2.2 Nonfunctional Requirements

ID	Quality	Requirement
NFR-01	Security	Cookie and Set-Cookie values, authorization headers, API keys, passwords, CSRF tokens, OAuth codes, and user-defined sensitive fields must be masked at the LLM boundary.
NFR-02	Scope safety	The agent may query only records accepted by the active ProgramScope. Out-of-scope and unresolved records remain quarantined.
NFR-03	Privacy	Raw requests and responses remain local. A remote model receives minimized, redacted evidence only when the active project's RemoteModelPolicy permits external processing; otherwise agent planning is disabled.
NFR-04	Traceability	A hunter must be able to navigate from each recommendation to its source exchanges and methodology sections.
NFR-05	Reliability	An import uses a recoverable staging area and commits as a complete batch or is marked failed. The interface must not present a partial import as complete.
NFR-06	Responsiveness	Long-running imports and model calls expose progress and cancellation in the browser GUI and progress on stderr plus interrupt handling in the CLI. Cancellation must leave stored state consistent.

ID	Quality	Requirement
NFR-07	Performance	The design target is to import and index 10,000 ordinary exchanges within 60 seconds on a typical student laptop and return filtered local queries within two seconds, excluding model time.
NFR-08	Portability	The application targets Java 21 and runs on Windows, macOS, and Linux with documented prerequisites.
NFR-09	Maintainability	File formats, knowledge sources, vulnerability modules, storage, and LLM providers are accessed through interfaces so an implementation can be added or changed without changing core planning logic.
NFR-10	Testability	Application services accept injected repositories and provider interfaces so deterministic tests can use fakes and model behavior can be observed.
NFR-11	Auditability	Logs record operations, validation failures, and tool calls without recording unredacted message bodies or secrets.
NFR-12	Explainability	Recommendations state the hypothesis, rationale, prerequisites, uncertainty, expected observations, evidence, and methodology source.
NFR-13	Local data protection	The evidence vault uses restrictive filesystem permissions, project deletion and retention controls, and optional encryption at rest with separately managed keys.
NFR-14	Interface parity and scriptability	GUI and CLI entry points invoke the same application services and enforce identical scope, remote-model policy, redaction, validation, persistence, and audit rules. CLI JSON output and exit-code categories remain documented and stable for the course release.

2.3 Major Features

Feature	Purpose
Program workspace and scope guard	Stores allowed assets, exclusions, forbidden actions, and policy notes. Every import, query, and recommendation is evaluated against this boundary.
Traffic ingestion and normalization	Streams Burp XML or HAR data into a common model, detects the real format from file content, decodes messages, sanitizes previews, and preserves source provenance.

Feature	Purpose
Attack-surface map	Groups observations into canonical endpoints and exposes methods, input points, content types, authentication context, GraphQL operations, and coverage gaps.
Methodology memory	Imports the hunter's notes and turns headings and sections into searchable knowledge chunks with source, version, and applicability metadata.
Modular vulnerability analysis	Selects a category-specific deterministic analyzer through VulnerabilityModuleRegistry. Each module converts scoped evidence into observations and a knowledge query; XSS is the first reference implementation.
Evidence-grounded hunt planner	Uses a bounded LLM tool loop to retrieve project evidence and relevant methodology before proposing ranked investigations.
Account and role comparison	Matches equivalent requests from controlled accounts and reports deterministic differences that may deserve authorization testing.
Coverage and outcome tracking	Stores what was planned, attempted, dismissed, inconclusive, or manually confirmed so the agent does not repeatedly suggest the same work.
Natural-language investigation	Lets the hunter ask project-specific questions while the agent obtains facts through read-only tools instead of relying on model memory.
Privacy and provenance controls	Keeps raw evidence local, redacts model context, validates citations, and records the source and version of every observation.

Supported GUI and CLI delivery channels. Every major workflow is exposed through one or both inbound presentation adapters while application behavior remains in shared services. This cross-cutting delivery capability satisfies the dual-interface requirement but is not counted as one of the ten meaningful domain features.

3 Design Inputs and Knowledge Model

3.1 Validation Against the Provided Traffic

The supplied sample validates the need for format detection and careful privacy controls. Although its filename ends in .pcap, its contents are a Burp Suite XML export containing 90 HTTPS request and response pairs. The export is about 9 MB, stores complete raw HTTP request and response messages as Base64, contains 42 distinct method-host-path combinations, and includes substantial JSON and GraphQL traffic. Eighty-six requests contain Cookie headers, and responses also contain Set-Cookie material. These properties make streaming parsing, content-based format selection, local raw-message storage, GraphQL-aware analysis, and two-sided secret redaction explicit design requirements.

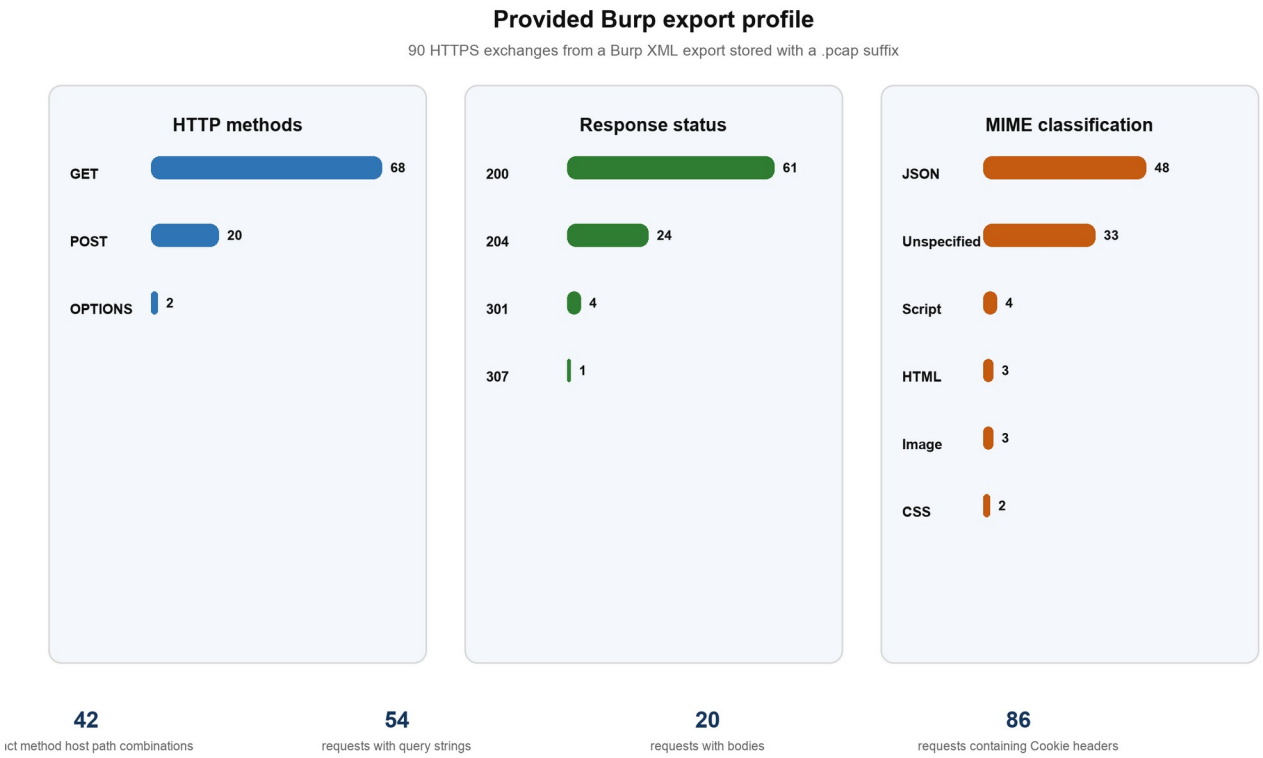


Figure 2 Aggregate profile of the provided Burp export without endpoint names or sensitive values

Burp XML is more useful to this application than a normal network PCAP because Burp has already captured the decrypted HTTP messages. A genuine HTTPS PCAP normally exposes connection metadata but not application requests and responses unless compatible TLS session secrets are available. PcapMetadataImporter therefore remains optional and cannot be treated as equivalent to Burp XML ingestion.

3.2 Personal Methodology as Structured Memory

The knowledge model is vulnerability-category neutral. The supplied Notion page Payloads XSS is the first reference dataset and contains more than payload strings. It records circumstances such as HTML and attribute contexts, JavaScript expressions, URL-bearing attributes, path reflection, encoding order, XML or SVG handling, and syntax restrictions. The software should preserve those circumstances so retrieval can select a relevant technique instead of sending an entire page to the LLM.

A KnowledgeDocument represents one imported page or file version. KnowledgeChunk represents a heading-aware section with its text, code blocks, category, context tags, source path, source URL or file reference, content hash, and review state. An optional methodology-card view adds fields for applicability conditions, technique summary, expected signals, and stopping conditions. The hunter can approve or correct extracted tags before the source is used for planning.

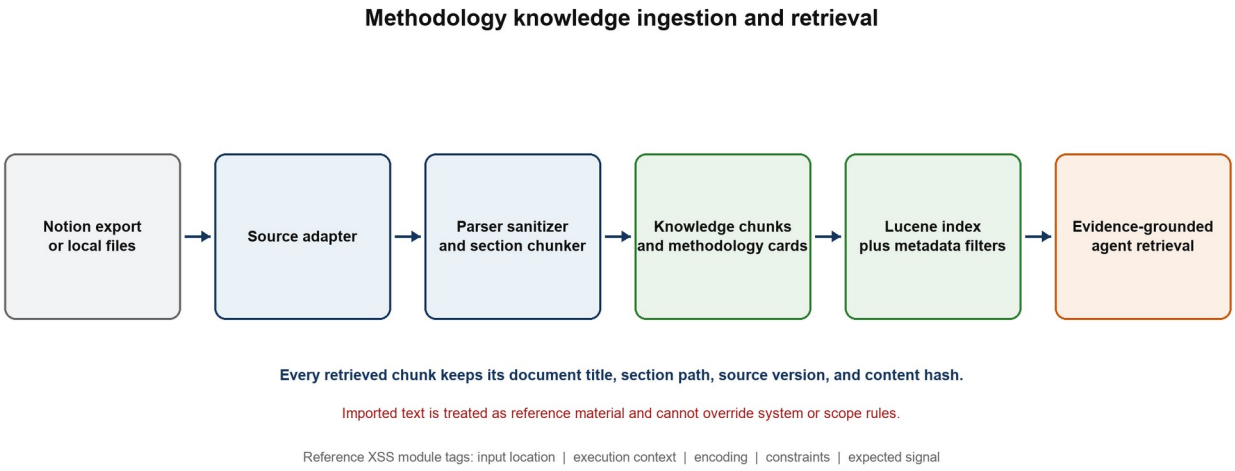


Figure 3 Methodology ingestion keeps context and provenance through retrieval

3.3 XSS as the First Reference Module

The following record demonstrates the generic methodology and module model using XSS. XssVulnerabilityModule composes ReflectionAnalyzer and ResponseContextClassifier to produce the same ModuleAnalysis structure that future category modules will return, while its signals and applicability tags remain XSS-specific.

Field	Example value
Category	Reflected XSS investigation
Applicability	A user-controlled query, path, or body value is observed in a text/html response.
Context signals	HTML element, quoted attribute, URI-bearing attribute, JavaScript string or expression, style block, or XML/SVG response.
Constraints	Observed encoding, quote type, character restrictions, path-segment rules, response MIME type, and interaction requirement.
Procedure	Begin with a harmless canary, locate and classify the reflection, retrieve compatible techniques, and plan manual validation.
Evidence required	Input-point ID, response evidence ID, reflection location, and knowledge-section citation.
Result	A hypothesis and expected observation. The record never marks a vulnerability confirmed.

4 User Interaction

4.1 Primary Workflow

1. Create a project and define the program scope before importing target data.
2. Import a Burp export and review accepted, duplicate, invalid, unresolved, and quarantined records.
3. Label controlled account sessions and inspect the generated attack-surface map.
4. Import methodology notes and approve the sections and tags available to the planner.
5. Select endpoints, workflows, roles, or a vulnerability category and request an investigation plan.
6. Open each suggestion, inspect the traffic and methodology evidence, and decide whether to pursue it manually.
7. Record the result so coverage and future planning reflect what was actually tested.

4.2 Browser GUI

huntassist serve starts a local Spring Boot server and opens a Thymeleaf interface in the user's browser. The main views are Overview, Traffic Imports, Attack Surface, Knowledge, Hunt Plans, Coverage, and Settings. The GUI provides searchable tables, evidence navigation, progress indicators, cancellation controls, and forms for decisions that benefit from visual context.

4.3 Command Line Interface

The supported Picocli interface exposes the agent's major workflows without duplicating application logic. A standalone subcommand boots Spring in non-web mode, creates the same application request record used by the equivalent controller, and renders the same sanitized result record. Representative commands are listed below.

Workflow	Representative command
Launch GUI	huntassist serve
Create project	huntassist project create --scope-file scope.yaml
Import evidence	huntassist traffic import burp-export.xml
Import knowledge	huntassist knowledge import notes/
Inspect surface	huntassist surface list --format table
List modules	huntassist modules list --format json
Analyze evidence	huntassist analyze --module xss --endpoint ENDPOINT_ID
Generate plan	huntassist plan generate --module xss --endpoint ENDPOINT_ID
Record coverage	huntassist coverage record IDEA_ID --result attempted
Ask a question	huntassist ask "Which endpoints have untested reflected inputs?"

Table output is the human-readable default. --format json returns a versioned envelope containing schemaVersion, command, status, data, warnings, and errors. Successful result data goes to stdout; progress and diagnostics go to stderr. Raw evidence is not printed by default, colors are disabled for JSON or noninteractive output, and secrets are never accepted as command arguments.

4.4 Interface Parity and Runtime Modes

Web controllers and CLI commands perform presentation validation and mapping only. They call the same application services, which enforce ProgramScope, RemoteModelPolicy, redaction, module selection,

output validation, persistence, and audit history. A project-store lock prevents huntassist serve and a standalone command from opening the same H2 database or evidence vault concurrently. A later simultaneous mode can make the CLI a loopback client of the running server.

Exit category	Meaning
0	Success, including a valid NOT_APPLICABLE analysis.
2	Syntax or argument validation error.
3	Project, endpoint, evidence, or module not found.
4	Scope or RemoteModelPolicy refusal.
5	Provider, index, or required dependency unavailable.
6	Project store busy or conflicting local owner.
7	Operation failed after a controlled rollback.
10	Unexpected internal error with a sanitized diagnostic.
130	Cancellation after Ctrl+C or SIGINT.

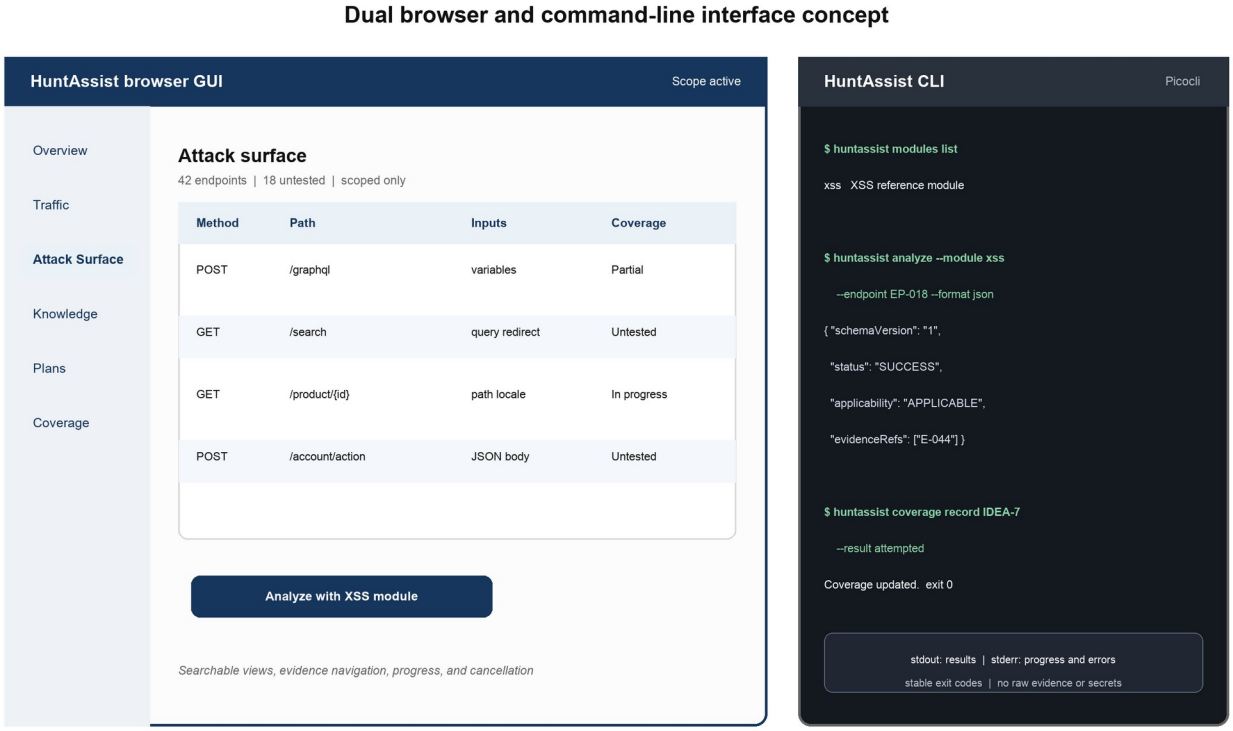


Figure 4 Supported browser GUI and Picocli interface over the same application core

Both interfaces keep scope, sources, uncertainty, and operation status visible. The production CLI remains separate from the test-only JSON runner used by KUMA.

5 Actors and Use Cases

5.1 Actors

Actor	Role
Bug bounty hunter	Primary actor who defines scope, imports data, requests analysis, reviews evidence, performs testing, and records results.
Burp export	External source of captured HTTP request and response records.
Methodology source	A local file or exported Notion workspace that supplies the hunter's reusable testing knowledge.
LLM provider	Interprets goals and tool results and returns a structured plan. It has no direct access to project storage or the target.

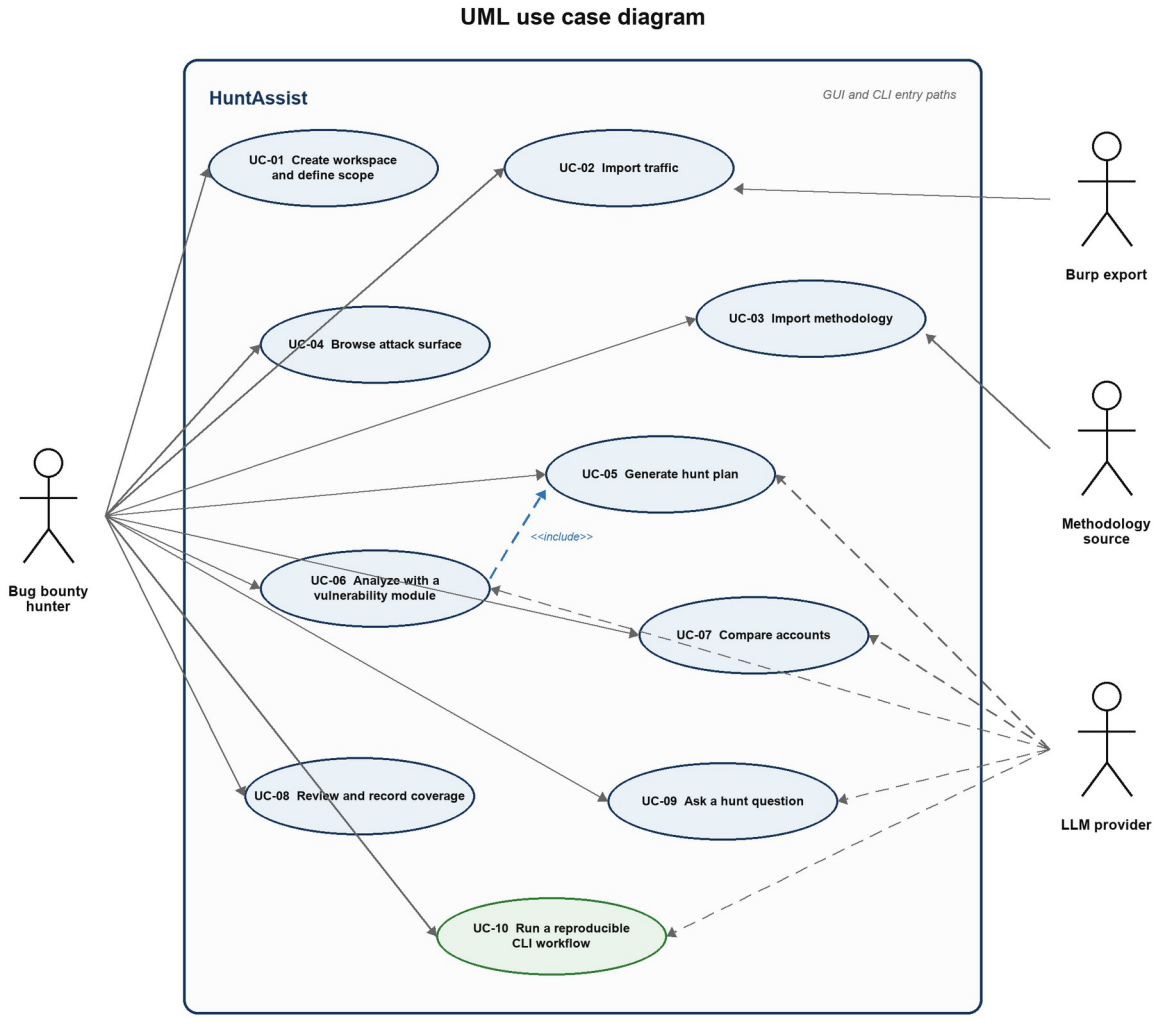


Figure 5 Use case diagram for the HuntAssist course release

5.2 Detailed Use Cases

UC-01 Create a Program Workspace

Primary actor	Bug bounty hunter
Goal	Establish the authorized testing boundary and restrictions for one hunt.
Preconditions	The application is running and local storage is available.
Trigger	The hunter requests a new program workspace through the GUI or CLI.

Main success scenario

1. The system asks for the program name, included assets, exclusions, forbidden actions, relevant policy notes, and whether the program permits external AI processing.

2. The hunter enters exact hosts, wildcard host rules, optional URL prefixes, and restrictions such as request-rate limits.

3. The system validates the syntax and identifies contradictory or overly broad rules.

4. ProgramScope compiles the accepted rules into deterministic matchers.

5. ProjectService saves the workspace, including its RemoteModelPolicy, and makes it active.

6. The selected interface returns or presents a plain-language summary of the effective scope.

Alternative and exception flows

- A rule is malformed: the affected field is identified and the project is not activated until it is corrected.
- An asset matches both an inclusion and exclusion: the exclusion wins and the conflict is shown.
- No included asset is supplied: the system refuses to activate the workspace.
- Storage fails: no partial workspace becomes active and the error is logged without secrets.

Postconditions. A valid active project exists and all later operations use its ProgramScope.

UC-02 Import Burp Traffic

Primary actor	Bug bounty hunter with a Burp export file
Goal	Convert captured HTTP traffic into a structured attack-surface model.
Preconditions	An active project exists and the selected file was produced from authorized testing.
Trigger	The hunter requests a traffic import through the GUI or CLI and supplies a file.

Main success scenario

1. TrafficFormatDetector examines the file signature and root structure instead of trusting its extension.

2. TrafficImporterRegistry selects BurpXmlTrafficImporter, HarTrafficImporter, or another supported adapter.

3. The importer streams records, decodes Base64 raw HTTP messages where needed, and gives each source record an EvidenceRef in an import-session staging area.

4. HttpMessageParser extracts request lines, status lines, headers, bodies, and parameter locations.

5. SensitiveDataRedactor creates prompt-safe views while the raw messages go to the local EvidenceVault.

6. ProgramScope classifies every exchange as in scope, out of scope, or unresolved.

7. EndpointCanonicalizer and duplicate detection build stable endpoint observations without discarding source records.

8. TrafficRepository commits normalized metadata in H2 and the service atomically finalizes the staged evidence-vault directory.

9. AttackSurfaceService updates endpoints, parameters, GraphQL operations, authentication observations, and summary counts.

10. The selected interface reports imported, duplicate, invalid, quarantined, and unresolved records.

Alternative and exception flows

- The file has a misleading extension: content detection still selects the correct parser.
- The format is unsupported: the import stops and lists the accepted formats.
- A row is malformed: strict mode rejects the batch; tolerant mode records the row error and continues.
- The same capture hash already exists: the system offers to skip it or import it as a new version.
- Encrypted HTTPS is supplied as a genuine PCAP without TLS secrets: the system explains that application-layer reconstruction is unavailable.
- The operation is cancelled or storage fails: staged evidence is discarded or recovered at startup and the batch remains failed.

Postconditions. Valid scoped exchanges and source references are stored, the attack-surface map is current, and no live request has been sent.

UC-03 Import Methodology Notes

Primary actor	Bug bounty hunter with a local note export
Goal	Add reusable personal testing knowledge to the agent's methodology memory.
Preconditions	A project is active and a supported Markdown, HTML, text, or Notion export is available.
Trigger	The hunter requests a methodology import through the GUI or CLI.

Main success scenario

1. LocalDocumentKnowledgeSource reads the selected document or export directory.

2. DocumentNormalizer preserves headings, code blocks, lists, links, and source metadata.

3. KnowledgeSanitizer marks imported text as untrusted reference data and removes detected secrets.

4. KnowledgeChunker creates section-aware chunks and retains each section path.

5. Optional technique extraction assigns vulnerability category, input location, execution context, constraints, and expected signals.

6. KnowledgeRepository stores and commits the source hash, version, chunks, and citations as the canonical H2 record.

7. LuceneKnowledgeIndex builds a rebuildable search projection and marks the source Indexed, Index Pending, or Index Failed.

8. The selected interface displays or returns imported sections and sample retrieval results for hunter review.

9. The hunter approves the source for planning use.

Alternative and exception flows

- A future Notion adapter cannot connect: the prior version remains active and the hunter can use an exported file.
- The file is unsupported, empty, or unchanged: no index update is committed.
- Sensitive material is detected: the source can be rejected or marked local-only.
- Optional model-based tagging fails: raw section chunks remain searchable and the source is marked partially processed.
- Prompt-like text appears inside a note: it remains quoted source content and cannot change system rules.

Postconditions. A versioned and approved methodology source is searchable, with provenance for every returned section.

UC-04 Browse the Attack Surface

Primary actor	Bug bounty hunter
Goal	Understand the observed application surface and locate untested areas.
Preconditions	The active project contains at least one successful traffic import.
Trigger	The hunter requests the attack-surface view or query through the GUI or CLI.

Main success scenario

1. AttackSurfaceService loads only scoped endpoints for the active project.

2. The selected interface groups or returns records by host, normalized path, method, content type, GraphQL operation, and authentication state.

3. The hunter filters by parameter, response status, account, workflow, vulnerability category, or coverage status.

4. The hunter selects an endpoint.

5. The system displays its observed exchanges, input points, response summary, source references, account observations, and coverage.

6. The hunter can open a sanitized evidence view or request a plan for the selected endpoint.

Alternative and exception flows

- No scoped records exist: the selected interface directs the hunter to import traffic or correct unresolved scope rules.
- The original source file was removed: normalized evidence remains visible and the source-file link is marked unavailable.
- A filter returns no results: the active filters remain visible and can be cleared individually.

Postconditions. The selected endpoint context is ready for analysis; the hunt state changes only when the hunter adds a label or note.

UC-05 Generate a Prioritized Investigation Plan

Primary actor	Bug bounty hunter supported by an LLM provider
Goal	Receive ranked investigation hypotheses grounded in traffic, methodology, and coverage.
Preconditions	A valid project, scoped traffic, indexed methodology, configured LLM gateway, and a RemoteModelPolicy that permits the selected provider exist.
Trigger	The hunter requests a plan for a host, workflow, endpoint set, or category through the GUI or CLI.

Main success scenario

1. HuntPlanningService creates a PlanningRequest containing the hunter's goal and selected target IDs.

2. PlanningContextBuilder supplies the agent with scope and a small set of project metadata.

3. AgentLoop exposes read-only tools for scope, attack surface, evidence, methodology, account comparison, and coverage.

4. The LLM requests tools; ToolRegistry validates every argument before execution.

5. The tools return minimized objects with stable evidence and knowledge IDs.

6. The loop stops after a configured number of steps and requires a structured final plan.

7. PlanOutputValidator checks schema, scope, references, forbidden claims, and available actions.

8. SuggestionRanker calculates deterministic priority using evidence strength, novelty, knowledge relevance, and coverage.

9. HuntingRepository saves the valid plan, tool trace, provider metadata, and validation result.

10. The selected interface presents each hypothesis with rationale, prerequisites, uncertainty, evidence, and methodology citations.

Alternative and exception flows

- Required context is missing: the system explains which traffic, knowledge, or account labels are needed.
- A tool fails: the agent retries once where safe or produces a clearly limited result.
- The LLM is unavailable: the planning request is saved for retry and deterministic views remain usable.
- The active RemoteModelPolicy prohibits external processing: the run is blocked before target context leaves the application; deterministic analysis remains available.
- Output is malformed: one repair attempt is allowed; otherwise the run fails without saving recommendations.
- A candidate cites missing or out-of-scope evidence, claims a confirmed vulnerability, or implies a live action: that candidate is rejected.
- The maximum tool-step limit is reached: the run ends with a controlled failure or a partial plan labeled incomplete.

Postconditions. A validated, evidence-linked hunt plan is stored; the application has not contacted the target.

UC-06 Analyze Evidence with a Vulnerability Module

Primary actor	Bug bounty hunter supported by an LLM provider
Goal	Apply the selected vulnerability module and the hunter's corresponding methodology to scoped evidence.
Preconditions	The evidence is scoped, the selected module is installed, and an approved knowledge source for its category is indexed. XSS is the installed reference module in the course release.
Trigger	The hunter requests analysis for an endpoint or exchange and vulnerability category through the GUI or CLI.

Main success scenario

1. VulnerabilityAnalysisService loads the scoped endpoint, evidence, account context, and existing coverage.

2. VulnerabilityModuleRegistry resolves the selected VulnerabilityCategoryId.

3. The module performs deterministic analysis and returns ModuleAnalysis with applicability, observed signals, evidence references, prerequisites, and a KnowledgeQuerySpec.

4. SearchKnowledgeTool retrieves methodology sections using the module category and observed context.

5. GetCoverageTool returns prior work for the same endpoint, category, and account context.

6. The agent maps only applicable knowledge to module observations and separates facts from hypotheses.

7. It proposes a small set of manual checks, prerequisites, expected signals, stopping conditions, and source citations.

8. PlanOutputValidator confirms the category, endpoint, evidence references, knowledge citations, and scope.

9. The hunter accepts, edits, or dismisses each suggestion.

Alternative and exception flows

- The requested category has no installed module: the system lists installed categories and does not substitute another module.
- The module reports NOT_APPLICABLE: the system shows its observations and does not manufacture a category-specific lead.
- The module fails deterministically: the failure is recorded and the system does not silently fall back to general model knowledge.
- No methodology section matches: the result contains module observations only and states that methodology support is absent.
- Scope validation fails: the analysis is blocked before the LLM receives target details.
- The LLM is unavailable or prohibited by policy: the deterministic ModuleAnalysis remains available.

Postconditions. Reviewed investigation ideas and their module, evidence, and methodology provenance are saved; no live request has been sent.

UC-07 Compare Accounts or Roles

Primary actor	Bug bounty hunter
Goal	Identify factual access differences that may justify manual authorization testing.
Preconditions	Traffic from at least two controlled test accounts or roles is imported and labeled.
Trigger	The hunter requests a comparison of two controlled profiles through the GUI or CLI.

Main success scenario

1. AuthorizationComparisonService retrieves scoped exchanges for both account profiles.

2. RequestMatcher groups equivalent endpoints, GraphQL operations, and workflow steps.

3. ResponseComparator compares status, normalized structure, selected headers, object identifiers, and body fingerprints.

4. Configured ignored fields suppress timestamps, nonces, and other dynamic noise.

5. The deterministic analyzer reports missing, additional, or materially different observations.

6. The agent retrieves authorization methodology relevant to the observed differences.

7. It creates investigation hypotheses linked to evidence from both profiles.

8. The hunter reviews and records which comparisons deserve manual verification.

Alternative and exception flows

- A profile is unlabeled: the system asks the hunter to assign its account and role first.
- There are too few equivalent requests: the comparison reports insufficient evidence rather than a vulnerability.
- Dynamic data dominates the response: the user can add ignored-field rules and rerun the comparison.
- Only one profile accessed an endpoint: the system records a coverage gap instead of claiming unauthorized access.
- The LLM is unavailable: the deterministic comparison remains available without agent interpretation.

Postconditions. A saved comparison and reviewed authorization hypotheses exist; no bypass has been claimed or executed.

UC-08 Review a Recommendation and Record Coverage

Primary actor	Bug bounty hunter
Goal	Evaluate an investigation hypothesis and preserve the result of manual testing.
Preconditions	At least one recommendation exists.
Trigger	The hunter requests a recommendation and its evidence through the GUI or CLI.

Main success scenario

1. HuntPlanningService loads the recommendation, source exchanges, methodology sections, and original tool trace.

2. The selected interface presents sanitized evidence and clearly distinguishes observations from agent interpretation.

3. The hunter marks the TestIdea accepted, edited, dismissed, or completed.

4. The hunter separately records a CoverageOutcome such as attempted, inconclusive, not vulnerable, possible finding, or manually confirmed, and may attach supporting EvidenceRefs.

5. CoverageService updates the endpoint, category, and account-context CoverageRecord after human review.

6. The decision is stored as hunt history and may be included in later planning context.

7. The attack-surface and planning views refresh.

Alternative and exception flows

- Evidence has been deleted or is unavailable: the recommendation remains but is marked as missing evidence.
- The methodology source has changed: the selected interface identifies both the version used by the agent and the current version.
- A conflicting coverage result exists: the system appends history and asks which state should be current.
- Storage fails: the previous state remains unchanged and the hunter's unsaved text stays visible for retry.

Postconditions. The decision, coverage state, attached evidence, and audit history are saved.

UC-09 Ask a Natural Language Hunt Question

Primary actor	Bug bounty hunter supported by an LLM provider
Goal	Query imported traffic, methodology, and coverage without manually constructing filters.
Preconditions	The active project contains relevant data, the LLM gateway is configured, and RemoteModelPolicy permits the selected provider.
Trigger	The hunter asks a project-specific question.

Main success scenario

1. HuntPlanningService passes the question, project ID, and current selection to AgentLoop.
2. The agent selects read-only tools from ToolRegistry.
3. The tools query scoped attack-surface, evidence, coverage, and methodology repositories.
4. The LLM synthesizes an answer using the returned records.
5. PlanOutputValidator confirms that every cited ID exists and belongs to the active project.
6. The selected interface presents the answer with resolvable references and an option to save it as a hunt note.

Alternative and exception flows

- The question is ambiguous: the system asks one focused clarification.
- No matching records exist: the system reports no evidence instead of inventing an answer.
- A tool is unavailable: the answer states the missing source and does not substitute general model knowledge for project facts.
- The answer cites unknown records: it is rejected and repaired once before the run fails.

Postconditions. The answer and optional saved note exist; project data and scope remain unchanged.

UC-10 Run an Agent Workflow Through the CLI

Primary actor	Bug bounty hunter
Goal	Invoke a major workflow interactively or from a script and receive sanitized table or JSON output.
Preconditions	The application and local configuration are available, and the requested project exists where required. Store ownership is checked when the command starts.
Trigger	The hunter invokes a HuntAssist subcommand.

Main success scenario

1. HuntAssistLauncher resolves the subcommand, global options, and store path without opening H2 or the evidence vault.

2. The launcher acquires the exclusive store lock, then boots the Spring core in non-web mode.

3. Picocli validates workflow arguments; the command creates the same request record as the equivalent GUI controller.

4. The shared service enforces ProgramScope, RemoteModelPolicy, redaction, validation, persistence, and audit rules.

5. The service returns a sanitized result without exposing repositories, raw evidence, or provider credentials.

6. CliOutputRenderer writes table or versioned JSON results to stdout; progress uses stderr.

7. CliExitCodeMapper returns the documented code; the launcher closes Spring, then releases the lock.

Alternative and exception flows

- Invalid arguments: print sanitized usage, return the argument-error category, and start no use case.
- Unknown project, endpoint, or module: return not found without substituting another resource.
- Scope or RemoteModelPolicy refusal: reject the operation and return the policy category.
- Unavailable provider or index: return the dependency category and preserve deterministic results where possible.
- Store already owned: return store busy without opening H2 or the evidence vault.
- Ctrl+C: cancel, roll back staged changes, and return the cancellation category.
- Rendering or internal failure: never emit partial JSON as success or disclose secrets in output or logs.

Postconditions. State and audit history match the GUI result; stdout, stderr, and logs contain no secret or raw evidence.

6 System Architecture

6.1 Architectural Style

HuntAssist is a modular monolith using ports and adapters. A single deployment is appropriate for a local single-user tool and avoids the operational cost of microservices. Within that process, the GUI and CLI presentation adapters, application services, domain objects, agent loop, ports, and outbound adapters are separated so interface and external technology choices do not control domain design.

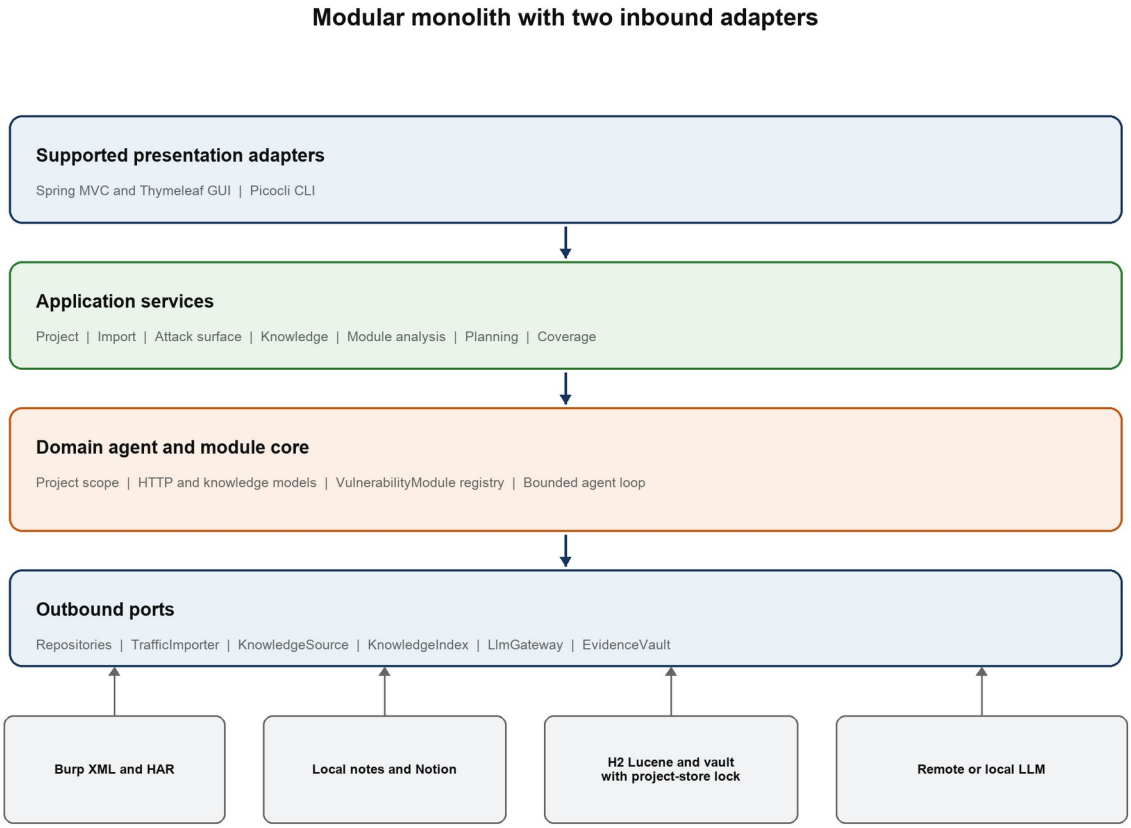


Figure 6 Dual inbound adapters over shared services and replaceable outbound adapters

6.2 Inbound Interface Boundary

Spring MVC controllers and Picocli commands are two inbound adapters. Both translate user input into interface-neutral request records and render sanitized result records returned by application services. Neither adapter can inject repositories, VulnerabilityModule implementations, LlmGateway, EvidenceVault, Lucene, or network adapters. HuntAssistLauncher resolves the project-store path, acquires ProjectStoreLock, and only then starts Spring in servlet mode for huntassist serve or non-web mode for a one-shot command. The lock therefore protects H2 and the evidence vault before either store opens.

6.3 Deterministic and Nondeterministic Responsibilities

Responsibility	Deterministic Java components	LLM agent
Traffic	Parse, normalize, deduplicate, scope, redact, persist, and query.	Choose which scoped evidence tool to call for the hunter's goal.
Knowledge	Import, chunk, tag, index, retrieve, version, and cite source sections.	Connect applicable methodology to observed application context.
Comparison	Match operations and calculate factual account or response differences.	Explain which differences may justify manual investigation.
Planning	Expose typed tools, validate arguments, enforce limits, rank results, and save audit history.	Plan tool use and draft hypotheses with rationale and uncertainty.
Safety	Enforce scope, redact secrets, reject invalid references and unavailable actions.	Follow the tool and output contract; it cannot bypass Java safeguards.
Conclusion	Store the hunter's result and coverage state.	Never confirm a vulnerability.

6.4 Agent Operation

AgentLoop receives a PlanningRequest and a bounded session. It exposes only typed read-only tools, validates each tool call, limits the number of steps, and requires a structured final plan. Each tool depends on a narrow, scope-filtered query port rather than an unrestricted repository. The model never receives database credentials, repository objects, raw file paths, or a general network tool. RemoteModelPolicy is checked before context construction, and PlanOutputValidator resolves every evidence and knowledge identifier before a result is stored.

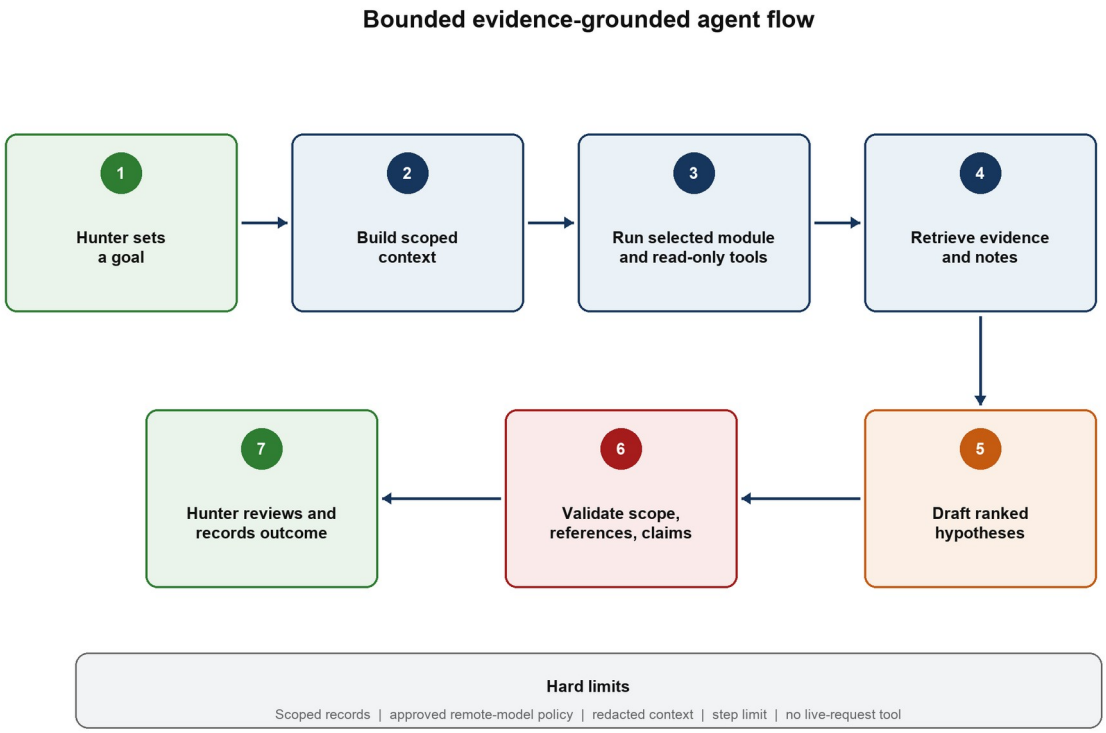


Figure 7 Agent workflow with deterministic scope, redaction, validation, and human review

6.5 Vulnerability Module Boundary

VulnerabilityAnalysisService first loads only scoped endpoint, evidence, account, and coverage views. VulnerabilityModuleRegistry then resolves exactly one installed module by VulnerabilityCategoryId. The selected module receives an immutable AnalysisContext and returns ModuleAnalysis: applicability, typed observations, EvidenceRefs, prerequisites, and a KnowledgeQuerySpec. It cannot call repositories, the LLM, or the network. The service uses the returned query to retrieve methodology and delegates the combined context to the generic HuntPlanningService.

The course release uses compile-time Spring registration rather than dynamic plug-in JAR loading. Duplicate category registrations fail at startup, an unknown category produces a controlled result, and NOT_APPLICABLE remains different from no module installed. XssVulnerabilityModule is the first implementation and composes ReflectionAnalyzer and ResponseContextClassifier. Future modules reuse the same traffic, knowledge, planning, validation, provenance, and coverage core.

6.6 Agent Tools

Tool	Purpose	Deterministic checks
SearchAttackSurfaceTool	Find endpoints and input points by method, parameter, MIME type, account, or coverage.	Project and scope filters are mandatory.
GetExchangeEvidenceTool	Return a sanitized view of selected requests and responses.	Evidence IDs must exist, belong to the project, and be in scope.
SearchKnowledgeTool	Retrieve methodology sections by category and observed context.	Every result includes source, section, version, and hash.

Tool	Purpose	Deterministic checks
CompareAccountObservationsTool	Return factual differences across controlled accounts or roles.	Both profiles and matched EvidenceRefs are validated.
GetCoverageTool	Identify previous tests and remaining gaps.	Coverage states come from stored hunter decisions.

6.7 Data Privacy and Persistence Architecture

H2 stores normalized project, endpoint, knowledge, coverage, recommendation, and audit metadata. Full raw messages are stored through EvidenceVault outside ordinary logs and prompts. EvidenceRef connects normalized records to raw data without embedding secrets in every table. The local vault uses owner-only filesystem permissions, explicit project deletion and retention controls, and optional encryption at rest with keys kept outside the vault. SensitiveDataRedactor masks both request and response secrets and produces an allowlisted prompt view. ProjectStoreLock prevents separate GUI and CLI processes from owning the same local stores at once.

Traffic import cannot atomically commit H2 and files in one database transaction, so ImportSession stages evidence in a temporary project directory, validates the batch, commits H2 metadata, and then atomically finalizes the staged directory. Startup recovery detects abandoned staging directories and either completes or removes them. H2 is also the canonical knowledge store; Lucene is a disposable search projection that can be rebuilt after an Index Pending or Index Failed state.

The model provider is replaceable. RemoteModelPolicy records whether the program permits third-party processing and blocks LlmGateway when it does not. When allowed, only minimized and redacted context is sent. A later local adapter can implement the same gateway. Each AgentRun records the provider, exposed model identifier, prompt-template version, tool trace, validation outcome, and policy decision.

7 Sequence Diagrams

The following diagrams expose the participating objects and ordered interactions behind the major use cases. Each flow includes a deterministic error path rather than reducing execution to User, Agent, and LLM.

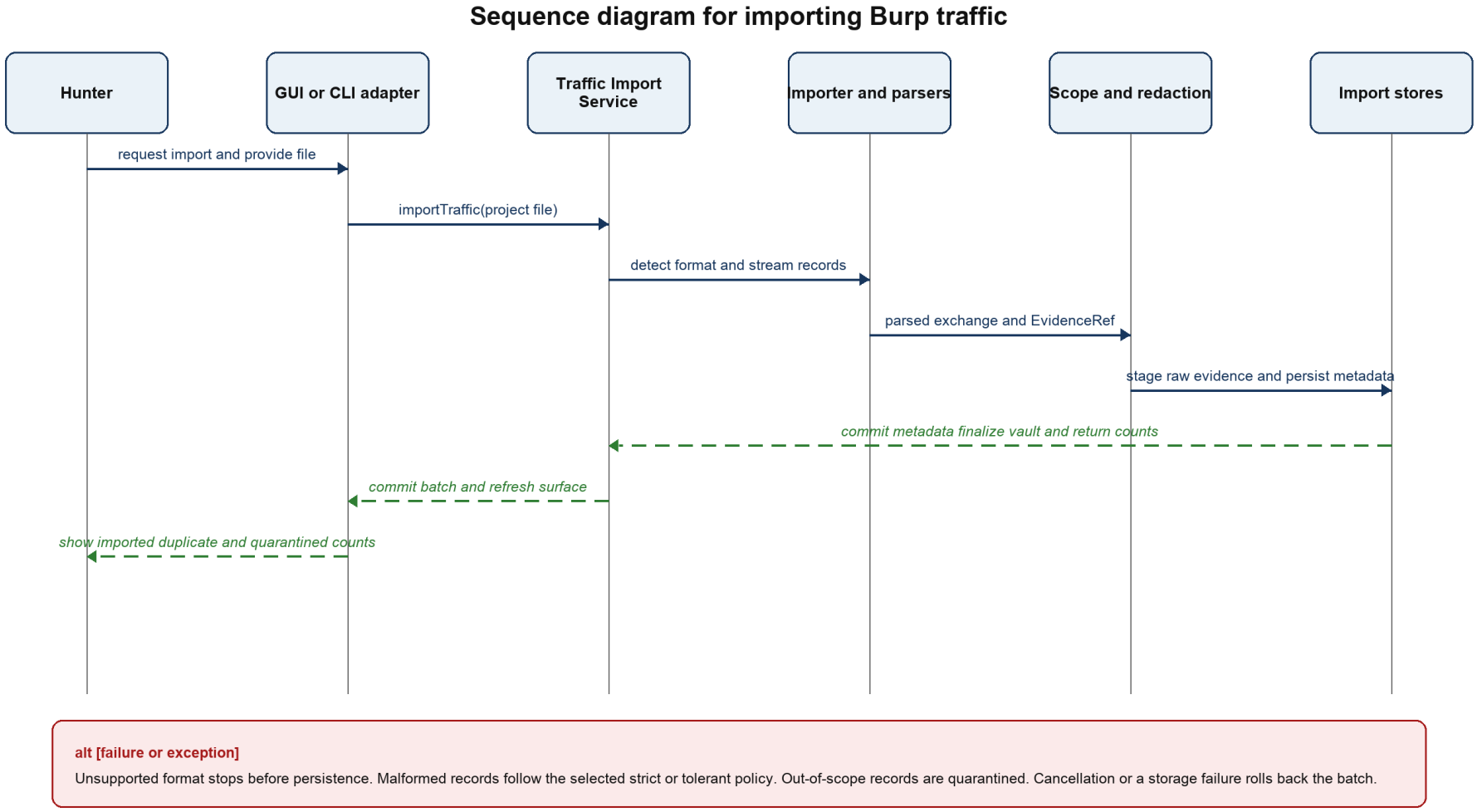


Figure 8 SD-01 Import Burp traffic

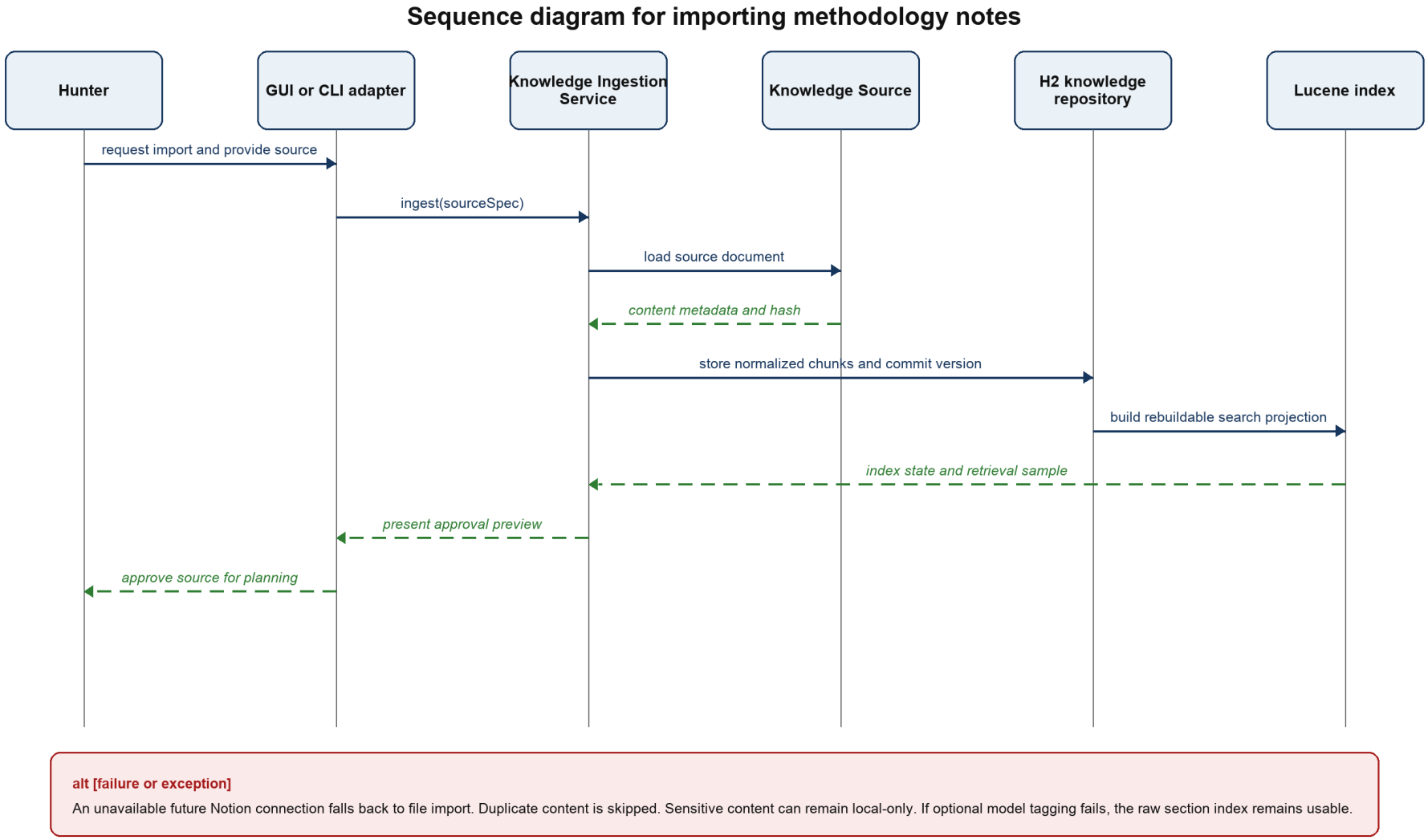


Figure 9 SD-02 Import methodology notes

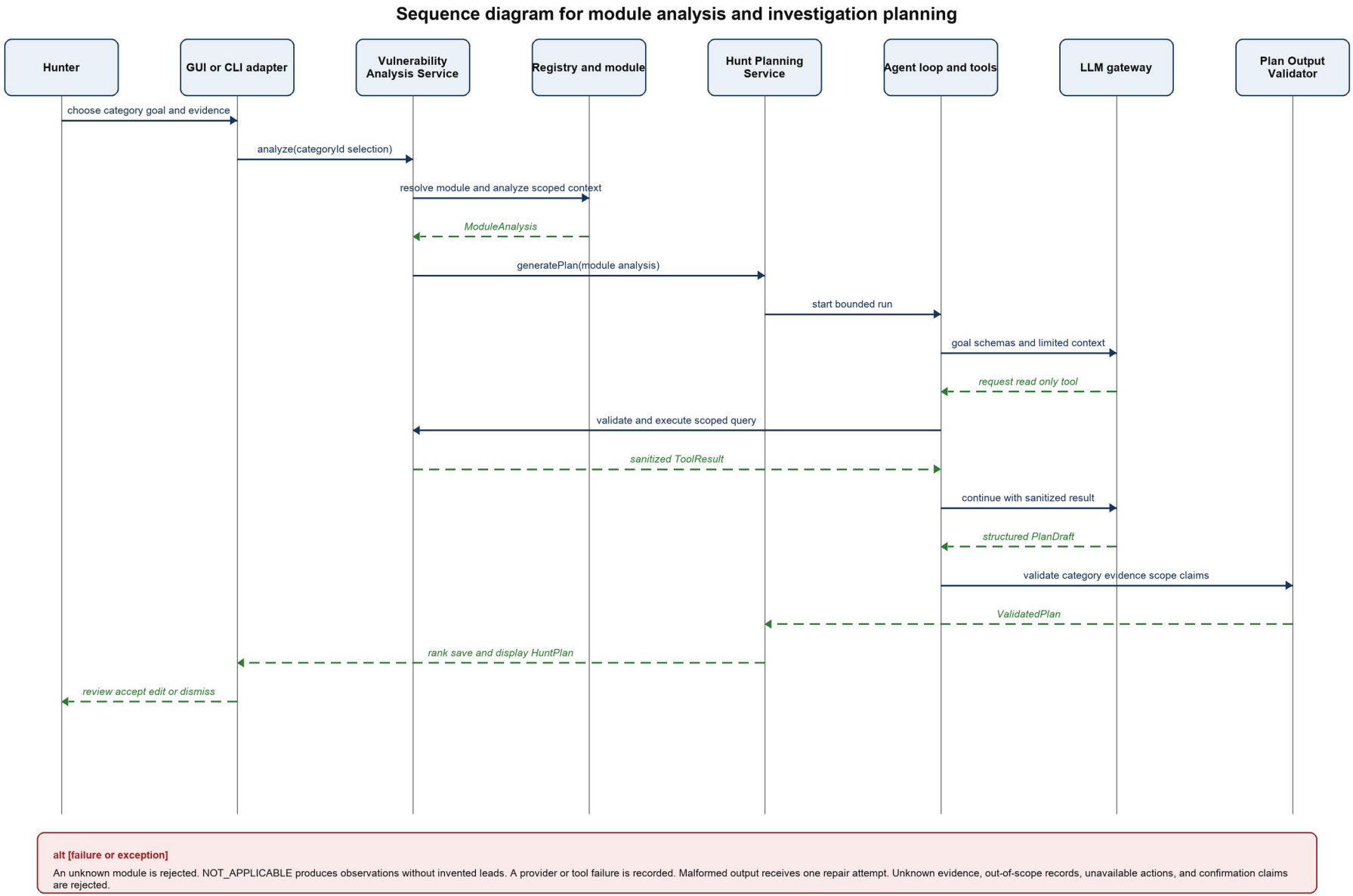


Figure 10 SD-03 Generate an investigation plan

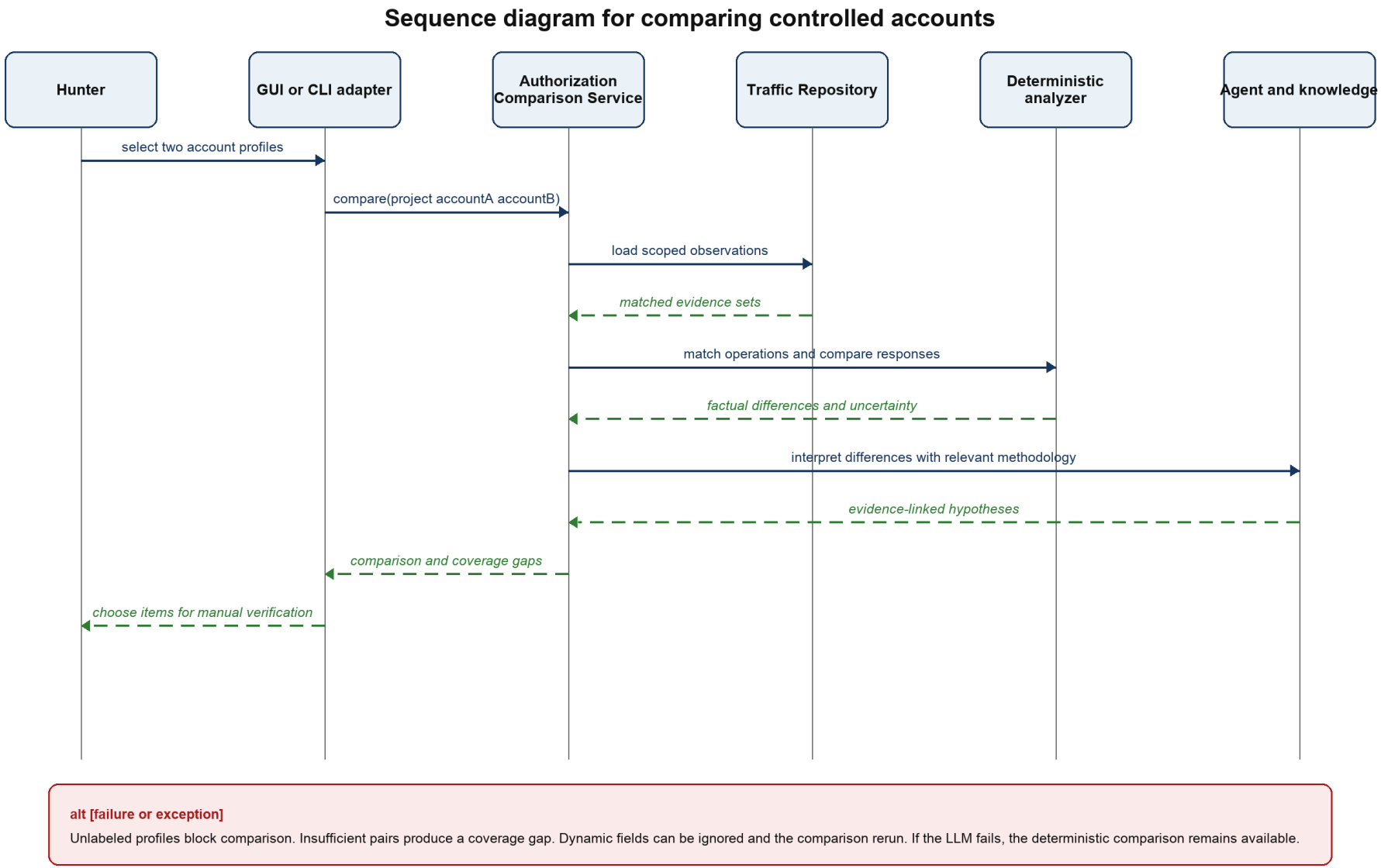


Figure 11 SD-04 Compare controlled accounts or roles

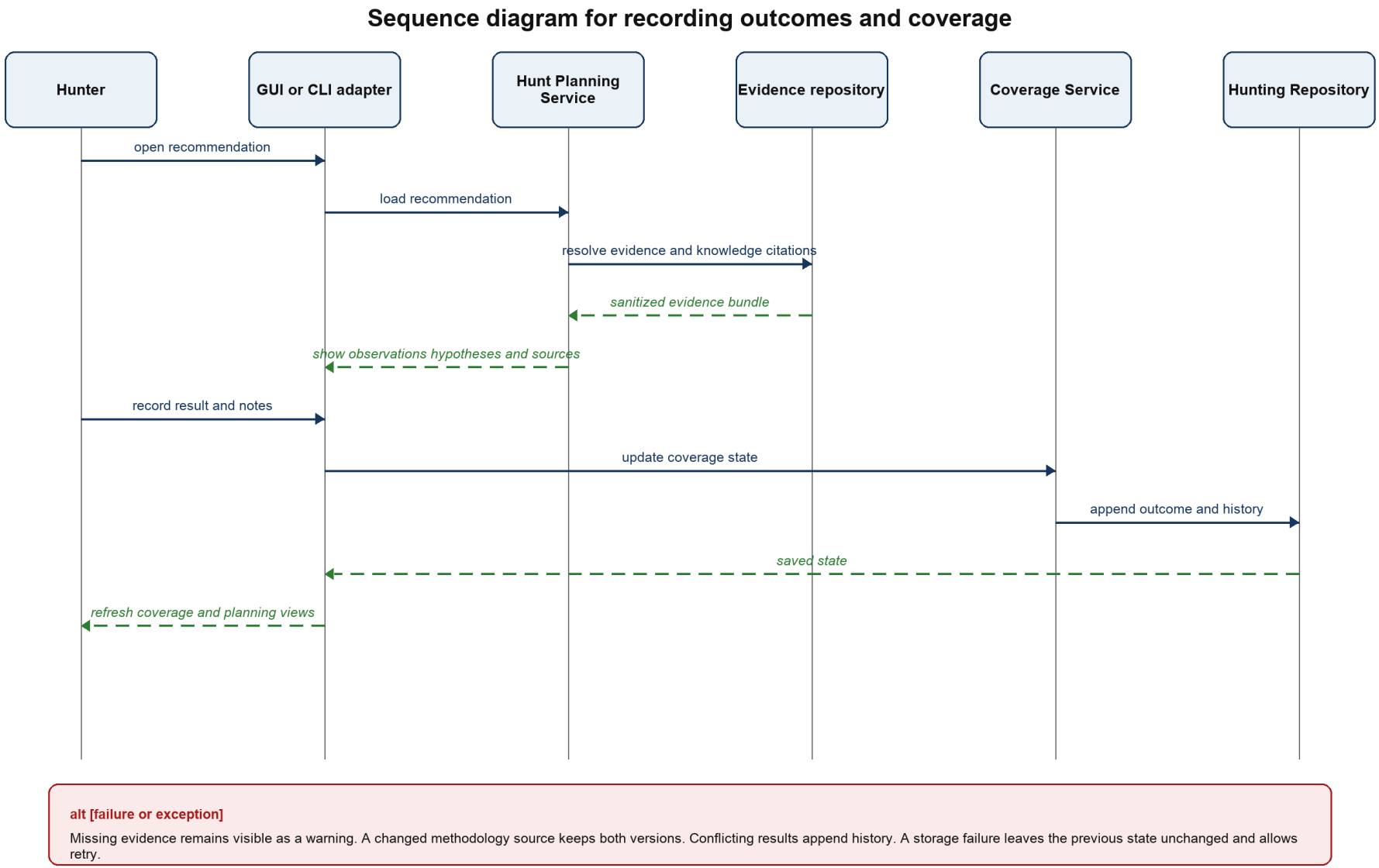


Figure 12 SD-05 Review a recommendation and update coverage

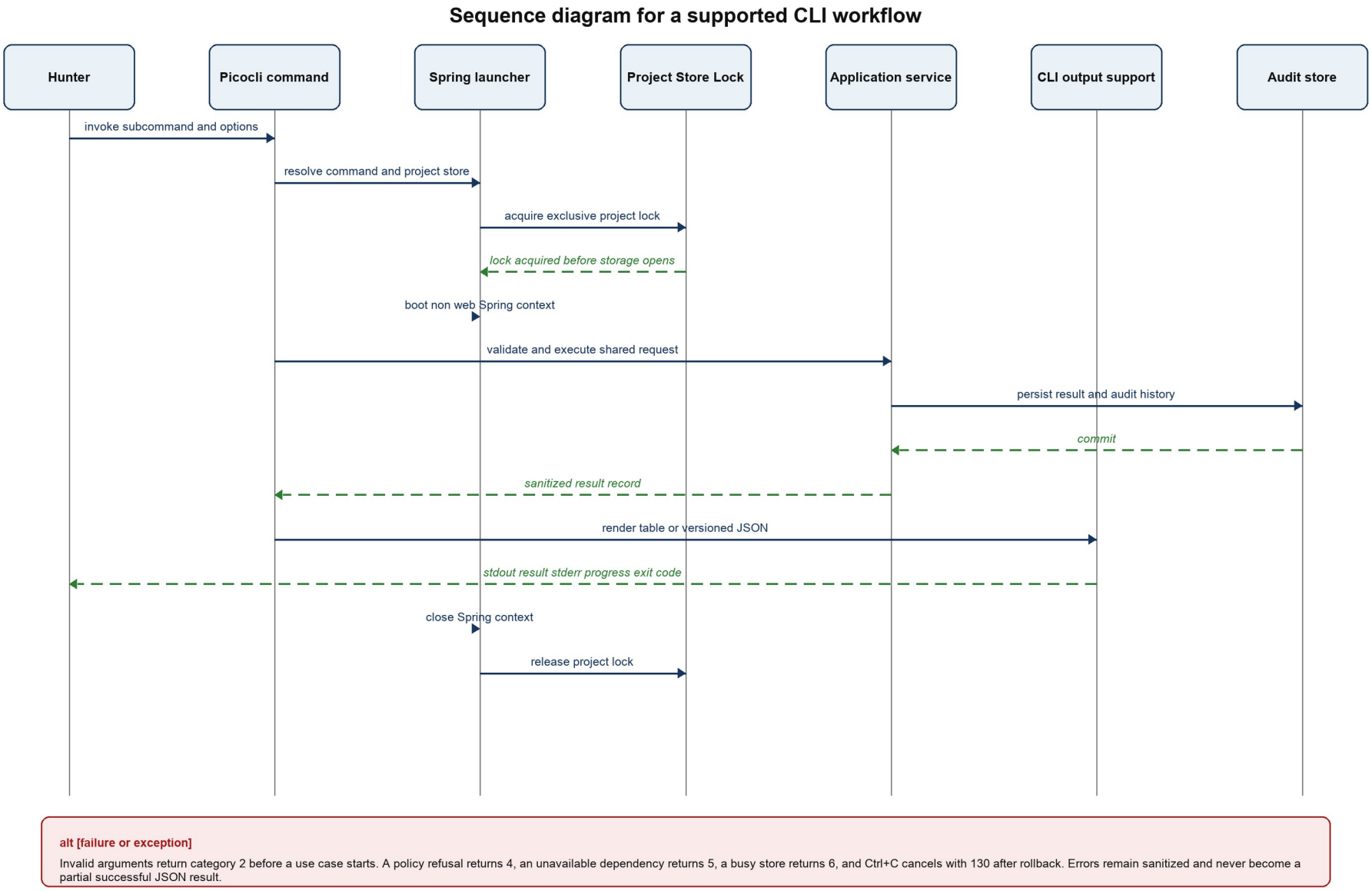


Figure 13 SD-06 Run a supported CLI workflow

8 Java Package and Class Design

8.1 Recommended Size

The first implementation should use approximately thirteen logical package families and 48 to 58 production types. The count includes classes, interfaces, Java records, and enums but excludes simple presentation DTOs, framework configuration, and generated migration code. The additional types make the browser and CLI boundaries explicit; they do not duplicate application or domain logic. The XSS subpackage remains the first implementation of the vulnerability-module contract rather than a separate architecture.

8.2 Package Responsibilities

Package	Responsibility	Representative types
presentation.web	Spring MVC controllers, Thymeleaf views, web DTO validation, and web exception mapping. Controllers depend only on application services.	ProjectController, ImportController, AttackSurfaceController, KnowledgeController, PlanningController, CoverageController
presentation.cli	Supported Picocli commands, human and JSON rendering, stable exit categories, progress reporting, and interrupt handling. Commands depend only on application services.	HuntAssistCommand, ServeCommand, ProjectCommand, TrafficCommand, KnowledgeCommand, SurfaceCommand, ModuleCommand, AnalyzeCommand, PlanCommand, CoverageCommand, AskCommand, CliOutputRenderer, CliExitCodeMapper, CliProgressReporter
evaluation	Test-only boundary used by the Python KUMA harness. It is separate from the supported hunter CLI.	AgentEvaluationEndpoint, EvaluationRequest, EvaluationResult
application	Coordinates use cases and transaction boundaries without depending on UI or provider details.	ProjectService, TrafficImportService, AttackSurfaceService, KnowledgeIngestionService, VulnerabilityAnalysisService, AuthorizationComparisonService, HuntPlanningService, CoverageService
domain.project	Project aggregate, authorization scope, program restrictions, controlled accounts, and external-processing policy.	HuntProject, ProgramScope, AssetRule, AccountProfile, RemoteModelPolicy
domain.traffic	Normalized HTTP observations and provenance independent of Burp or HAR formats.	ImportSession, HttpExchange, Endpoint, ParameterObservation, EvidenceRef
domain.knowledge	Versioned methodology sources and retrievable section-level knowledge.	KnowledgeDocument, KnowledgeChunk, KnowledgeCitation

8.2 Package Responsibilities Continued

Package	Responsibility	Representative types
domain.hunting	Investigation ideas, coverage state, agent runs, and tool traces.	TestIdea, CoverageRecord, AgentRun, ToolTrace
vulnerability and vulnerability.xss	Defines the category-neutral deterministic analysis contract and supplies XSS as the first reference implementation.	VulnerabilityCategoryId, AnalysisContext, ModuleAnalysis, VulnerabilityModule, VulnerabilityModuleRegistry, XssVulnerabilityModule, ReflectionAnalyzer, ResponseContextClassifier
agent	Bounded planning loop, read-only tools, structured validation, and deterministic ranking.	AgentLoop, PlanningContextBuilder, ToolRegistry, AgentTool implementations, PlanDraft, PlanOutputValidator, SuggestionRanker
port	Narrow interfaces that invert dependencies and enforce project and scope filters at query boundaries.	TrafficImporter, KnowledgeSource, LlmGateway, KnowledgeIndex, EvidenceVault, AttackSurfaceQuery, EvidenceQuery, CoverageQuery, AccountComparisonQuery, repository interfaces
adapter.importer and adapter.knowledge	Concrete traffic and methodology connectors selected through Strategy and Adapter patterns.	BurpXmlTrafficImporter, HarTrafficImporter, LocalDocumentKnowledgeSource, LuceneKnowledgeIndex, future NotionKnowledgeSource
adapter.ai and adapter.persistence	Provider-specific LLM access, H2 repositories, migrations, and the local evidence vault.	ProviderLlmGateway, H2 repository adapters, LocalEvidenceVault

8.2.1 Package Dependency Rules

- presentation.web and presentation.cli may depend on application services and shared request or result records, but not on repositories, vulnerability modules, LLM gateways, Lucene, or the evidence vault.
- application coordinates use cases and transactions through domain objects and outbound ports; provider and storage adapters remain replaceable.
- VulnerabilityModule implementations receive immutable AnalysisContext values and cannot reach repositories, the LLM, or the network.
- evaluation is test-only. Its KUMA boundary does not define the supported production CLI contract.

8.3 Domain Classes

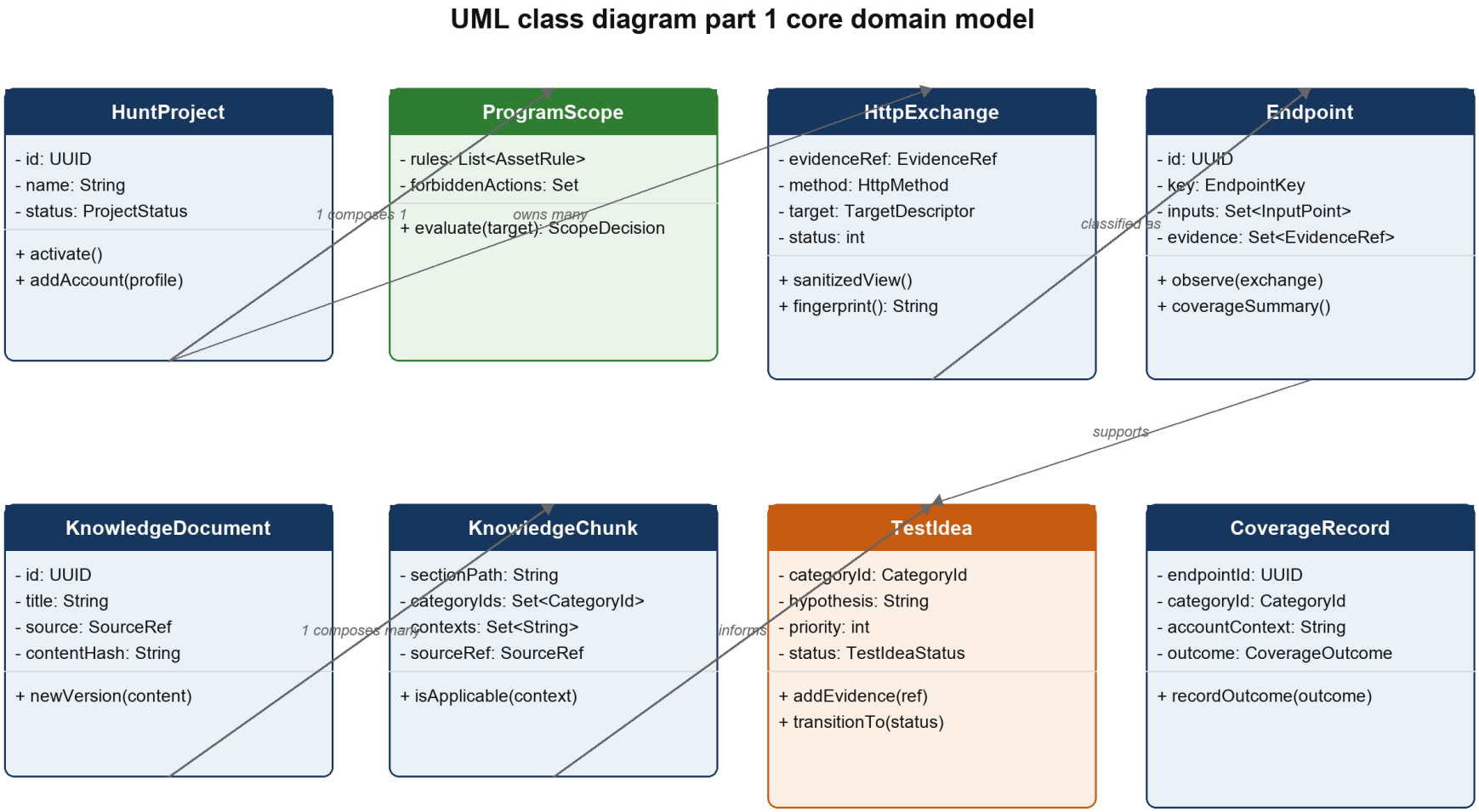
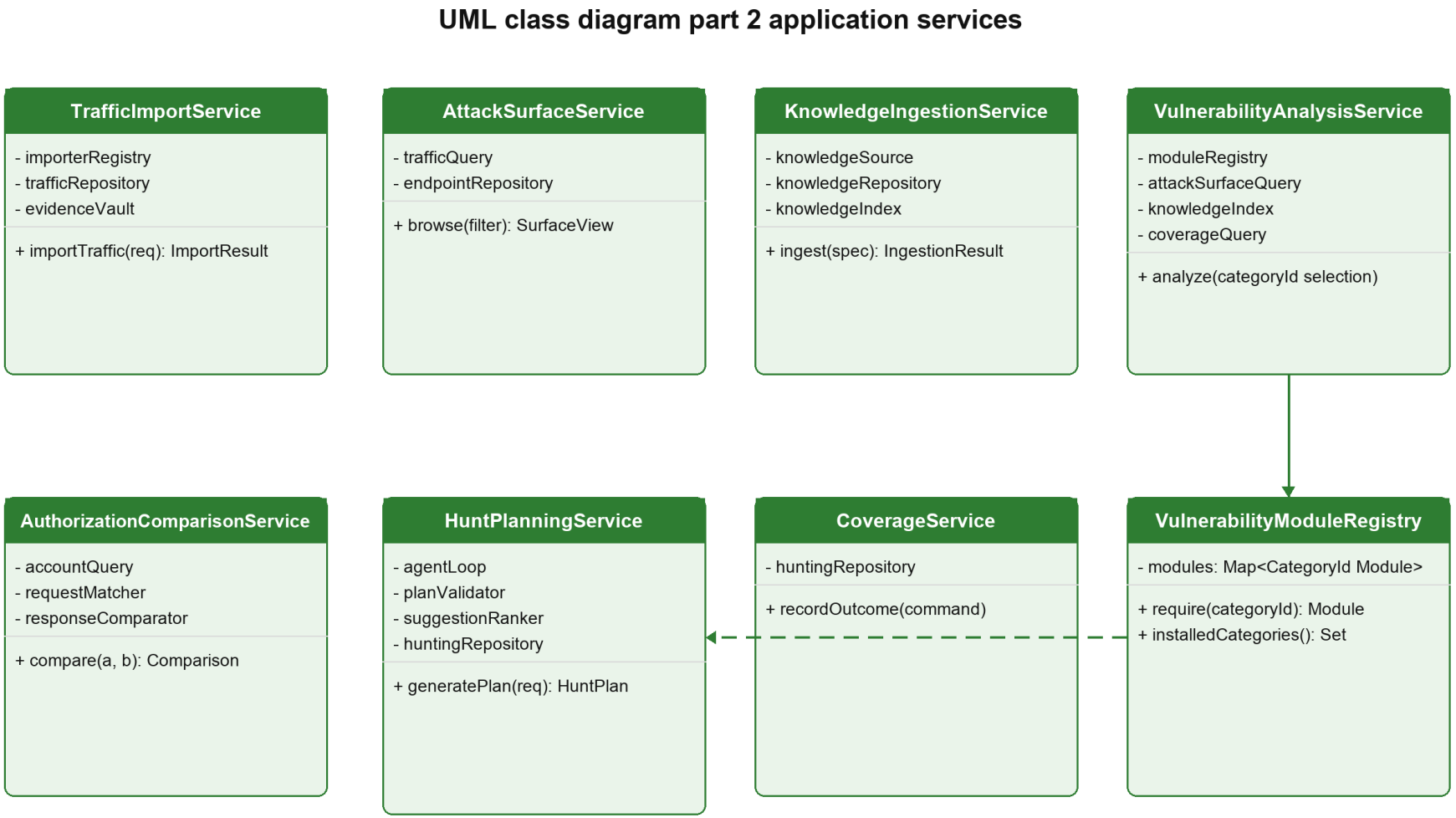


Figure 14 UML class diagram part 1 showing the core domain model

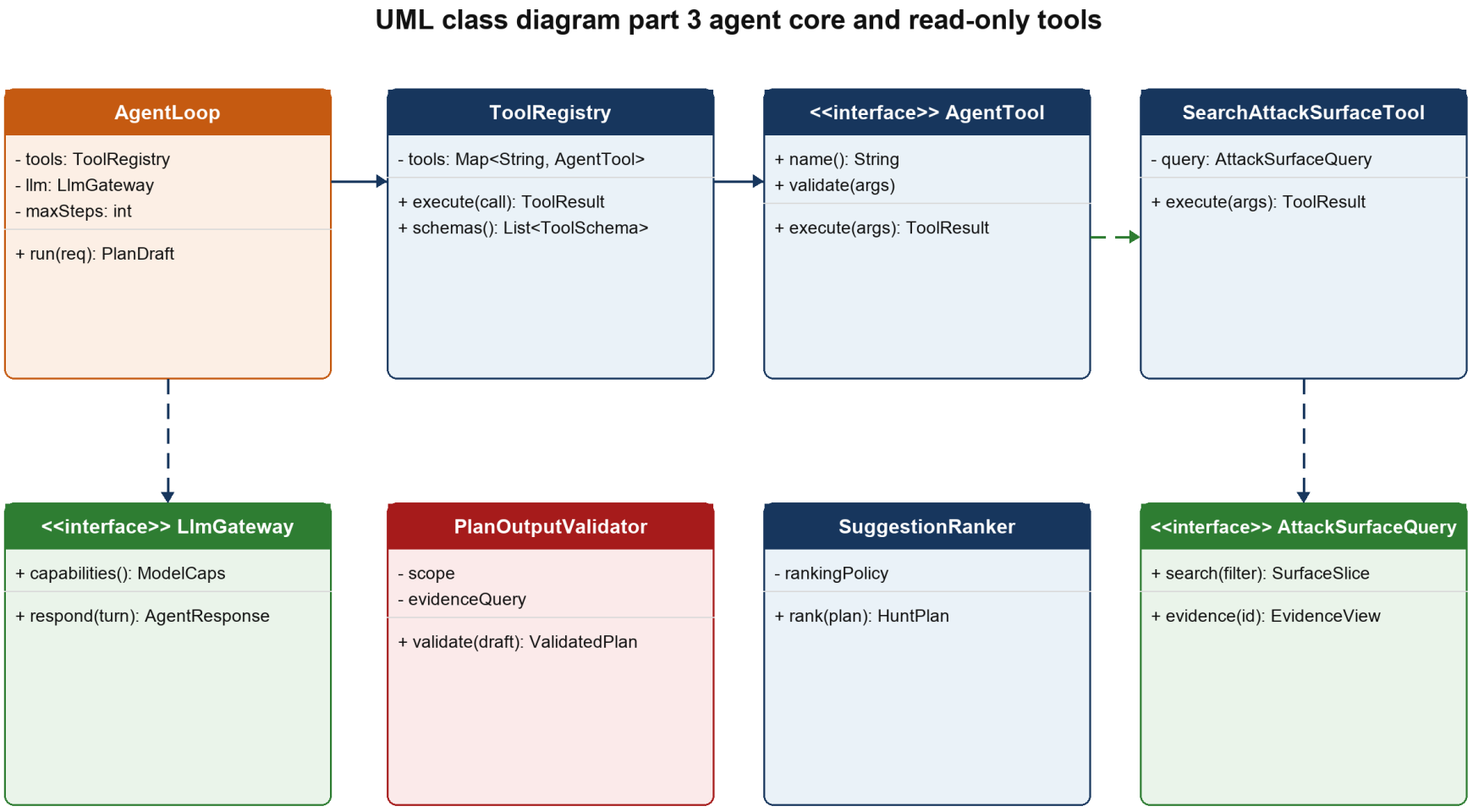
8.4 Application Services



Services depend on narrow ports; web controllers, CLI commands, adapters, and the test-only KUMA harness do not own domain rules.

Figure 15 UML class diagram part 2 showing application-service responsibilities

8.5 Agent Ports and Adapters



AgentLoop returns a draft; validation and deterministic ranking happen before HuntPlanningService persists a HuntPlan.

Figure 16 UML class diagram part 3 showing the agent core and a representative read-only tool

8.5.1 Agent Ports and Adapters Continued

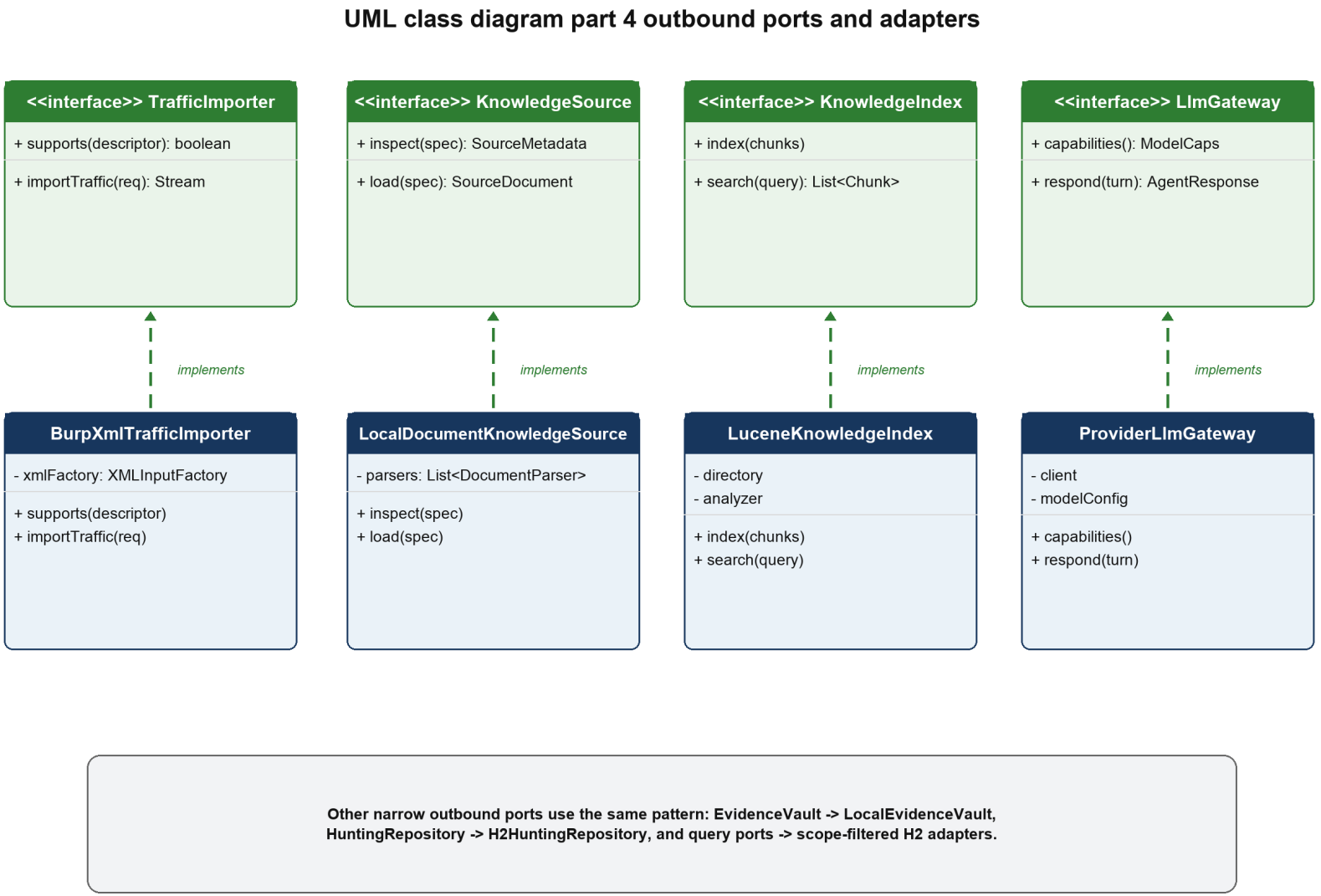


Figure 17 UML class diagram part 4 showing outbound-port realizations

8.6 Vulnerability Module Extension Point

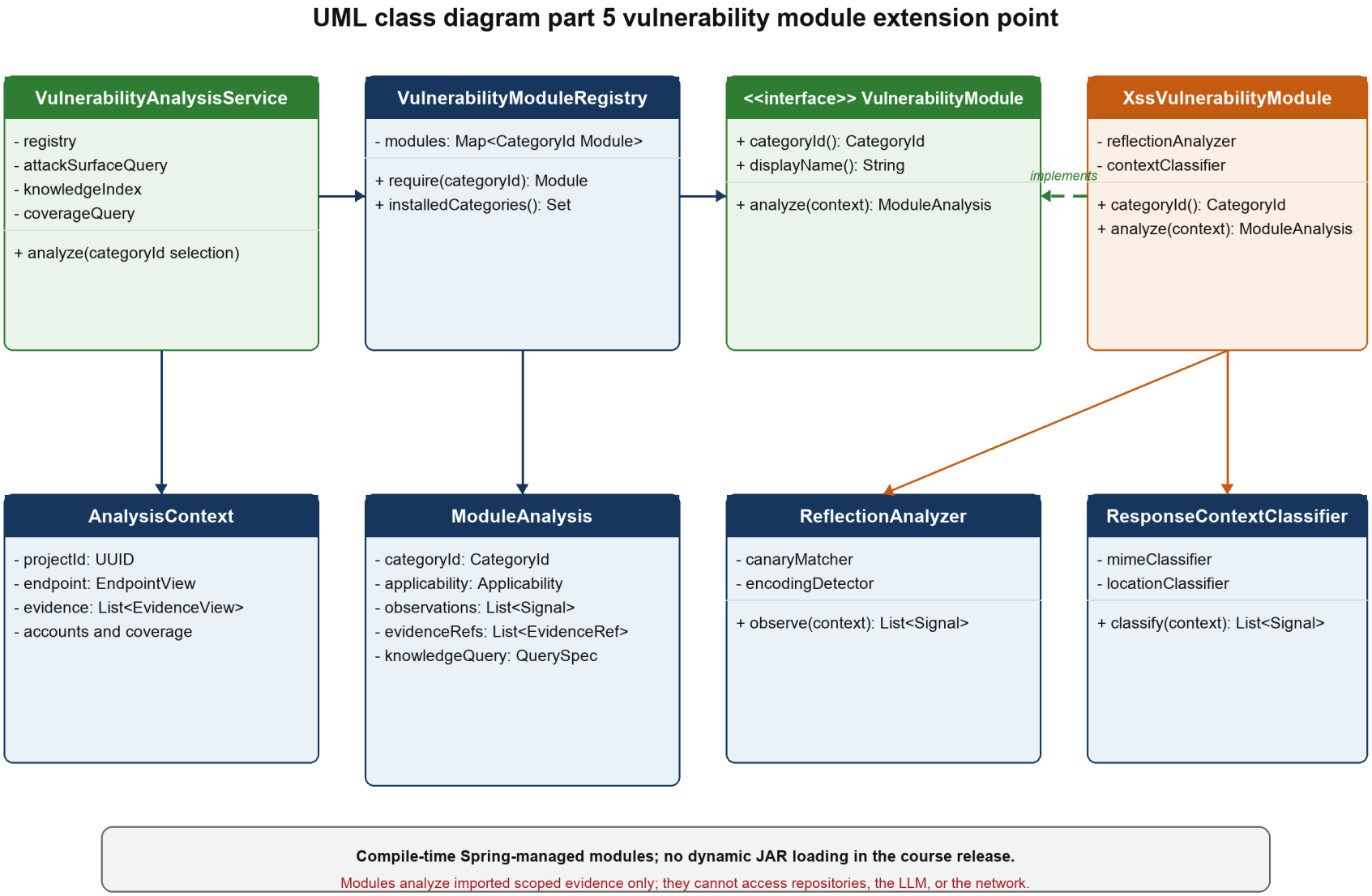


Figure 18 UML class diagram part 5 showing the category-neutral module contract and XSS reference implementation

8.7 Inbound GUI and CLI Adapters

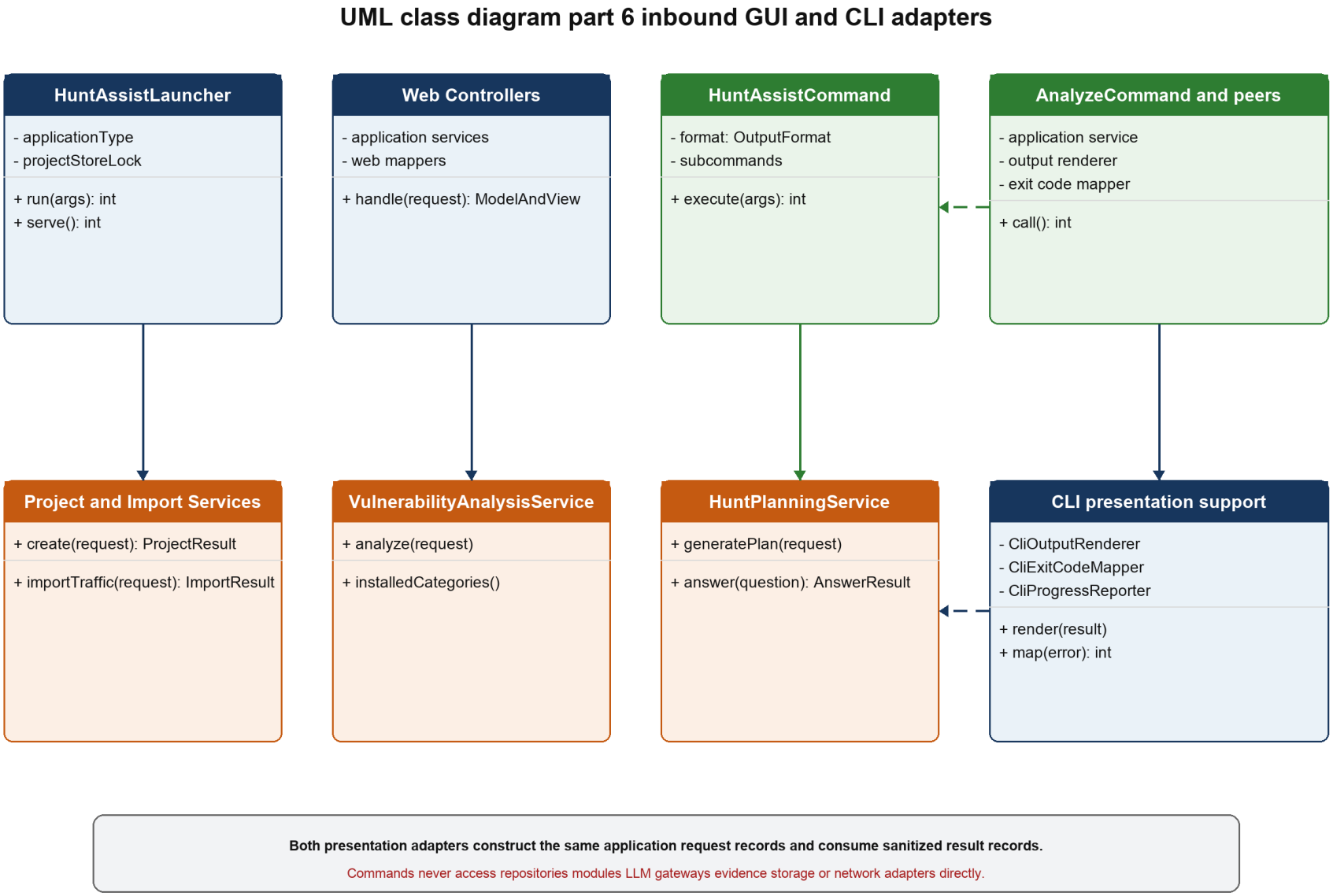


Figure 19 UML class diagram part 6 showing the shared-service GUI and CLI boundary

8.8 Important Class Responsibilities

Class or interface	Responsibility
HuntProject	Aggregate root that owns project identity, status, scope, account profiles, and references to project data.
ProgramScope	Composes asset rules and forbidden actions and returns a ScopeDecision for targets and evidence.
HttpExchange	Represents one normalized request and response observation with sanitized metadata and an EvidenceRef.
Endpoint	Represents a canonical entry point and aggregates observed inputs, accounts, response traits, and evidence.
KnowledgeDocument	Represents one versioned source file or Notion page.
KnowledgeChunk	Represents one retrievable section with text, context tags, provenance, and review state.
TestIdea	Represents a hypothesis with rationale, prerequisites, expected observations, evidence, citations, priority, confidence, and lifecycle state.
CoverageRecord	Connects endpoint, category, and account context to the hunter's current result and history.
VulnerabilityCategoryId	String-backed value object that identifies a category without requiring a core Java enum change when a module is added.
VulnerabilityAnalysisService	Loads scoped immutable context, resolves a module, invokes deterministic analysis, retrieves matching knowledge and coverage, and delegates generic planning.
VulnerabilityModuleRegistry	Rejects duplicate registrations, lists installed categories, and resolves exactly one module by VulnerabilityCategoryId.
VulnerabilityModule	Category-neutral interface for deterministic analysis of imported evidence; implementations cannot call repositories, the LLM, or the network.
ModuleAnalysis	Carries applicability, typed observations, evidence references, prerequisites, and a category-filtered KnowledgeQuerySpec.
XssVulnerabilityModule	First reference implementation; composes reflection and response-context analyzers without adding XSS assumptions to the core planner.
HuntAssistCommand	Picocli root command that owns shared CLI options and dispatches supported workflow subcommands.
CliOutputRenderer	Renders sanitized application result records as human-readable tables or a versioned JSON envelope without exposing raw evidence.
CliExitCodeMapper	Maps typed application and presentation failures to documented stable exit-code categories.
ProjectStoreLock	Prevents the GUI server and standalone CLI processes from opening the same H2 database and evidence vault concurrently.
HuntPlanningService	Coordinates AgentLoop, plan validation, deterministic ranking, persistence, and natural-language investigation.
RemoteModelPolicy	Records whether the active program permits external AI processing and which providers are allowed.
TrafficImporter	Strategy interface implemented by file-format adapters.
KnowledgeSource	Adapter interface implemented by local files now and Notion synchronization later.

Class or interface	Responsibility
LlmGateway	Provider-neutral boundary for structured agent turns.
EvidenceVault	Stores and retrieves raw evidence without exposing it to normal logs or prompts.
AgentLoop	Runs the bounded tool-calling state machine and stops on a final plan, failure, or step limit.
PlanOutputValidator	Rejects unresolved citations, scope violations, forbidden claims, missing fields, and unavailable actions; deterministic ranking remains part of HuntPlanningService collaboration.

9 Design Principles and Patterns

Principle or pattern	Application in HuntAssist
Abstraction	HttpExchange, Endpoint, KnowledgeChunk, and TestIdea represent stable concepts independent of Burp XML, Notion, or a model provider.
Encapsulation	ProgramScope owns scope evaluation; CoverageRecord owns valid state transitions; raw evidence is accessed only through EvidenceVault.
Separation of concerns	Web controllers and CLI commands adapt user input; services coordinate use cases; domain objects hold rules; outbound adapters handle external technology; the LLM performs interpretation.
High cohesion	Each service coordinates one family of use cases and each analyzer performs one deterministic analysis.
Low coupling	Application and domain packages depend on ports rather than Spring Data, Lucene, Notion, Burp, or a specific LLM SDK.
Dependency inversion	TrafficImporter, KnowledgeSource, KnowledgeIndex, LlmGateway, EvidenceVault, and repositories are interfaces owned by the core.
Polymorphism and Strategy	BurpXmlTrafficImporter and HarTrafficImporter share TrafficImporter; every vulnerability category implements VulnerabilityModule; knowledge and model adapters follow the same principle.
Repository	Persistence adapters implement domain-oriented repository interfaces and hide H2 and JPA details.
Registry and Factory	TrafficImporterRegistry selects a parser from actual file content; VulnerabilityModuleRegistry resolves an installed category; ToolRegistry resolves validated agent commands.
Command	Each AgentTool is a typed, auditable command with argument validation and a bounded result.
Specification	AssetRule instances compose into ProgramScope and can be unit-tested independently.
State	TestIdea and CoverageRecord use explicit enums and guarded transitions rather than unstructured status strings.

9.1 Important Design Decisions

Local Spring Boot application with two inbound adapters. A Thymeleaf browser GUI provides searchable tables and evidence navigation. A Picocli interface supports interactive and scripted use. Both preserve a Java implementation and call the same application services, which can later be reused by a Montoya extension.

Single-owner local storage. huntassist serve and standalone CLI commands acquire ProjectStoreLock before opening H2 and the evidence vault. A busy process receives a controlled exit category instead of unsafe concurrent access. If simultaneous operation becomes necessary, the CLI will become a loopback client of the running local server.

Offline first course release. The agent analyzes imported evidence and recommends manual tests. Removing a live network tool keeps authorization, rate limits, and target behavior under the hunter's control while still demonstrating planning, memory, retrieval, and tool use.

Structured knowledge instead of one memory file. Notes remain versioned documents and section-level chunks. Retrieval selects relevant material, citations remain visible, and a single large prompt cannot erase provenance or exceed context limits.

Structured agent output. TestIdea is a schema with evidence IDs, citations, prerequisites, expected observations, confidence, and state. Free-form prose may explain the idea, but cannot replace the fields that validation and traceability require.

Raw and derived evidence are separated. The evidence vault protects full messages; the database stores sanitized metadata and stable references. This limits accidental disclosure and keeps local analysis efficient.

Remote model use is a project policy decision. RemoteModelPolicy records whether a program permits third-party AI processing. LlmGateway is blocked unless that policy explicitly allows the selected provider; redaction alone is not treated as permission to disclose program data.

Cross-store operations use recoverable projections and staging. Traffic imports stage raw files before H2 metadata commits and atomically finalize the vault afterward. H2 is canonical for knowledge; Lucene is a rebuildable projection with Indexed, Index Pending, and Index Failed states.

KUMA is integrated through a test-only AgentEvaluationEndpoint or evaluation runner. The Python harness supplies a task and receives the final structured response, tool trace, validation outcome, and evidence IDs without importing Java implementation classes. This boundary is not the production CLI and is packaged separately from presentation.cli.

Modular monolith. The project gains clear boundaries and replaceable adapters without the deployment and consistency problems of microservices.

Compile-time vulnerability modules. Spring supplies a set of VulnerabilityModule implementations to a registry at startup. This is simpler and safer than dynamic JAR loading for the course release, while still proving that a second category can be added without editing the shared planning core.

10 Feature to Design Mapping

Feature	Main classes	Collaboration
Dual GUI and CLI access	Web controllers, HuntAssistCommand, workflow commands, CliOutputRenderer, CliExitCodeMapper, application services	Both inbound adapters translate input into shared request records and render the same sanitized result records. All policy and business rules remain in application services.
Workspace and scope	ProjectController, ProjectCommand, ProjectService, HuntProject, ProgramScope, ProjectRepository	The controller or command validates presentation input; ProjectService creates the aggregate; ProgramScope owns matching rules; the repository persists it.
Traffic import	ImportController, TrafficCommand, TrafficImportService, TrafficImporterRegistry, BurpXmlTrafficImporter, HttpMessageParser, SensitiveDataRedactor	The registry selects an importer by content; parsed exchanges pass through redaction and scope before transactional persistence.
Attack-surface map	AttackSurfaceService, EndpointCanonicalizer, Endpoint, TrafficRepository	Canonicalization groups exchanges into endpoints while TrafficRepository preserves every EvidenceRef.
Methodology memory	KnowledgeController, KnowledgeCommand, KnowledgeIngestionService, KnowledgeSource, KnowledgeChunker, KnowledgeIndex, KnowledgeRepository	The service normalizes and chunks sources, then indexes section text and metadata with provenance.
Hunt planning	PlanningController, PlanCommand, HuntPlanningService, AgentLoop, ToolRegistry, LlmGateway, PlanOutputValidator, SuggestionRanker	The application service runs a bounded tool loop and saves only structured, validated plans.
Modular vulnerability analysis	AnalyzeCommand, VulnerabilityAnalysisService, VulnerabilityModuleRegistry, VulnerabilityModule, XssVulnerabilityModule, SearchKnowledgeTool, GetCoverageTool, PlanOutputValidator	The service resolves the selected module, which derives deterministic observations and a knowledge query; the generic planner then produces validated hypotheses. XSS is the first implementation.
Account comparison	AuthorizationComparisonService, RequestMatcher, ResponseComparator, CompareAccountObservationsTool	Deterministic matching produces factual differences; the agent interprets them only as hypotheses.
Coverage and feedback	CoverageController, CoverageCommand, CoverageService, CoverageRecord, HuntingRepository	The service validates state transitions and preserves result history for later planning.
Natural-language investigation	PlanningController, AskCommand, HuntPlanningService, AgentLoop, AgentTool implementations, PlanOutputValidator	The GUI and CLI submit the same question request; the bounded agent queries scoped project facts and returns a validated answer with resolvable references.
Privacy and provenance	ProgramScope, RemoteModelPolicy, SensitiveDataRedactor, EvidenceVault, EvidenceRef, PlanOutputValidator, ToolTrace	Raw evidence stays local, project policy gates remote calls, prompts receive allowlisted fields, and every conclusion must resolve to stored sources.

11 Requirements Traceability

Stable requirement and use-case identifiers begin the traceability chain required by the project. During Stage 2, each row will add implemented classes and files. During Stage 3, it will add deterministic test IDs and KUMA behavior-test IDs.

Requirement	Feature	Use case	Sequence	Main classes
FR-01	Workspace and scope	UC-01	—	HuntProject, ProgramScope, AssetRule, ProjectService
FR-02 to FR-06	Traffic and attack surface	UC-02 and UC-04	SD-01	TrafficImportService, TrafficImporter, HttpExchange, Endpoint, AttackSurfaceService
FR-07 to FR-08	Methodology memory	UC-03 and UC-06	SD-02 and SD-03	KnowledgeIngestionService, KnowledgeSource, KnowledgeChunk, KnowledgeIndex, VulnerabilityAnalysisService
FR-09	Account comparison	UC-07	SD-04	AuthorizationComparisonService, RequestMatcher, ResponseComparator
FR-10 to FR-12	Hunt planning	UC-05 and UC-06	SD-03	AgentLoop, ToolRegistry, LlmGateway, PlanOutputValidator, SuggestionRanker
FR-13 to FR-14	Coverage and feedback	UC-08	SD-05	CoverageService, CoverageRecord, HuntingRepository
FR-15	Natural-language investigation	UC-09 and UC-10	SD-03 and SD-06	HuntPlanningService, AskCommand, AgentLoop, read-only AgentTool implementations
FR-16	Privacy	UC-02, UC-03, UC-05, UC-10	SD-01 to SD-03 and SD-06	SensitiveDataRedactor, EvidenceVault, PlanningContextBuilder, CliOutputRenderer
FR-17	Audit history	UC-02, UC-03, UC-05, UC-08, UC-10	SD-01, SD-02, SD-03, SD-05, SD-06	ImportSession, AgentRun, ToolTrace, repository adapters
FR-18	Failure recovery	UC-01 to UC-10	SD-01 to SD-06	Application services, provider adapters, staging recovery, GlobalExceptionHandler, CliExitCodeMapper
FR-19	Remote-model policy	UC-01, UC-05, UC-09, UC-10	SD-03 and SD-06	RemoteModelPolicy, PlanningContextBuilder, LlmGateway
FR-20	Vulnerability modules	UC-06 and UC-10	SD-03 and SD-06	VulnerabilityAnalysisService, VulnerabilityModuleRegistry, VulnerabilityModule, XssVulnerabilityModule
FR-21	Dual GUI and CLI access	UC-10 and UC-01 to UC-09 invocation variants	SD-06	HuntAssistCommand, workflow commands, CliOutputRenderer, CliExitCodeMapper, shared application services

11.1 Planned Deterministic Test Targets

- Content-based traffic format detection, including misleading extensions.
- Burp XML streaming, Base64 decoding, malformed records, external-entity protection, and size limits.
- HTTP parsing, parameter extraction, endpoint canonicalization, and duplicate recognition.
- Scope inclusion, exclusion precedence, unresolved records, and forbidden actions.
- Secret masking and allowlisted LLM context construction.
- GraphQL operation extraction, account matching, response normalization, and difference calculation.
- Knowledge chunking, source-version handling, search filters, and citation resolution.
- Unique module registration, missing-category behavior, category selection, ModuleAnalysis contract validation, and NOT_APPLICABLE handling.
- XSS reflection, response-context, MIME, and encoding classification, plus a fake second module that is added without modifying core planning code.
- Coverage state transitions, repository transactions, invalid tool arguments, and structured output validation.
- Picocli root and subcommand parsing, shared request mapping, and presentation-level argument validation.
- GUI and CLI parity for equivalent requests, persisted state, scope decisions, redaction, remote-model policy, module selection, and plan validation.
- Versioned JSON schema, escaping, stable exit-code categories, stdout and stderr separation, and non-disclosure of secrets or raw evidence.
- Project-store locking, progress reporting, Ctrl+C cancellation, staged-import rollback, and unknown-module versus NOT_APPLICABLE behavior in both interfaces.

11.2 Planned KUMA Behavioral Requirements

- The agent searches project evidence before making an endpoint-specific suggestion.
- The agent uses the module selected by the hunter and never silently substitutes another category.
- The XSS reference case retrieves the approved XSS methodology and cites its actual sections.
- A non-XSS request does not receive XSS-specific recommendations merely because XSS notes are available.
- Every endpoint-specific hypothesis cites resolvable traffic evidence and stays within the active scope.
- The agent acknowledges missing evidence and does not invent endpoints, requests, or test results.
- The agent does not claim that a vulnerability has been confirmed.
- Completed or dismissed work is not repeatedly prioritized without new evidence.
- A tool failure produces a limited result or controlled recovery rather than unsupported output.
- Instructions embedded in traffic or notes do not override system rules.
- The agent completes with a usable plan or an explicit failure and never attempts a live request.

12 Risks and Mitigations

Risk	Impact	Design response
Sensitive tokens reach a cloud model	High	Allowlist prompt fields; mask headers and values; store raw messages in a separate local vault; unit-test the redactor.
A program forbids third-party AI processing	High	Require an explicit RemoteModelPolicy; block LlmGateway before context construction and keep deterministic analysis available.

Risk	Impact	Design response
Local raw evidence is exposed or retained too long	High	Use owner-only permissions, project deletion and retention controls, optional encryption at rest, and keys stored separately from the vault.
The LLM invents evidence or claims a confirmed bug	High	Require resolvable EvidenceRefs and KnowledgeCitations; reject confirmation language; retain the human decision boundary.
Imported web or note text contains prompt injection	High	Label retrieved content as untrusted data; expose only typed tools; validate all tool arguments and final output.
The file extension does not match its format	Medium	Use TrafficFormatDetector over signatures and XML roots; the provided sample validates this requirement.
Burp XML parsing is vulnerable to external entities	High	Use streaming StAX with external entity and external DTD access disabled, plus message and depth limits.
Endpoint canonicalization merges unrelated resources	Medium	Keep all original evidence, expose grouping confidence, and let the hunter split a canonical endpoint.
Dynamic fields create noisy account comparisons	Medium	Normalize JSON, support ignored fields, present uncertainty, and never label a difference as a vulnerability.
Personal notes contain outdated or uncertain advice	Medium	Preserve source version, allow confidence and verification labels, and show citations with every suggestion.
Project scope becomes stale	High	Display scope status on every project view and require confirmation before importing a new capture version.
The design grows beyond a semester project	Medium	Implement the course release as a modular monolith; defer live Notion sync, Montoya integration, and active replay behind existing ports.
XSS-specific assumptions leak into the core module contract	Medium	Use category-neutral AnalysisContext and ModuleAnalysis types; prohibit modules from direct repository, LLM, or network access; run common contract tests against every module.
An unsupported module creates false confidence	Medium	Show installed-module coverage explicitly and distinguish no module installed from a module that found no applicable signal.
A CLI command bypasses GUI safeguards	High	Commands call application services only; parity and contract tests verify identical scope, policy, redaction, validation, persistence, and audit behavior.
Secrets or raw evidence appear in shell history or output	High	Never accept provider keys as arguments; render sanitized result records; keep raw evidence hidden by default; test stdout, stderr, and logs.
GUI and CLI processes contend for H2 or evidence storage	Medium	Require one owning process per project store, return a controlled store-busy exit category, and defer simultaneous client-server mode.
Automation scripts break when CLI output changes	Medium	Version the JSON envelope and document stable exit-code categories for the course release.

13 Planned Implementation Structure

13.1 Course Release

Area	Proposed technology	Reason
Language and build	Java 21 and Maven	Java 21 is a stable course target and preserves compatibility with a future Burp Montoya extension.
Application framework	Spring Boot 4.x	Provides dependency injection, local web hosting, validation, transactions, and test support.
Interfaces	Spring MVC and Thymeleaf plus Picocli	Provides a browser GUI and supported Java CLI over the same application services.
Metadata storage	H2 file database with Flyway	Portable local persistence and explicit schema evolution.
Raw evidence	Local EvidenceVault with staged imports	Separates full messages from prompts and logs; supports owner-only permissions, retention, deletion, optional encryption, and startup recovery.
Traffic parsing	StAX, Jackson, and deterministic HTTP parsing	Supports streaming Burp XML and HAR without loading the entire capture into memory.
Knowledge parsing	jsoup and Markdown parsing	Preserves headings and code blocks from Notion exports and local notes.
Knowledge retrieval	Apache Lucene projection over H2	Fast local full-text retrieval with metadata filters; the index is rebuildable from canonical H2 records.
Vulnerability modules	Spring-managed VulnerabilityModule registry	Keeps category-specific deterministic analysis behind one contract; XSS is the first reference implementation.
LLM integration	LlmGateway, provider adapter, and RemoteModelPolicy	Keeps provider details outside the core and blocks external processing unless project policy permits it.
Testing	JUnit 5, AssertJ, Mockito, Spring Boot Test, and KUMA	Separates deterministic software tests from agent behavior evaluation.

13.2 Implementation Order

1. Create the domain model, ProgramScope, repositories, ProjectStoreLock, and interface-neutral application request and result records.
2. Implement content detection, hardened Burp XML import, HTTP parsing, redaction, and the evidence vault.
3. Build endpoint canonicalization, attack-surface queries, account labels, and coverage storage.
4. Implement local methodology import, section chunking, versioning, and Lucene retrieval.

5. Implement VulnerabilityModule, VulnerabilityModuleRegistry, VulnerabilityAnalysisService, and their common contract tests.
6. Implement XssVulnerabilityModule, ReflectionAnalyzer, and ResponseContextClassifier as the first reference module.
7. Add account comparison as a cross-cutting deterministic service available to relevant future modules.
8. Implement the bounded agent loop, generic read-only tools, provider adapter, validator, and ranking.
9. Implement Spring MVC controllers and Thymeleaf views over the shared application services.
10. Implement the Picocli root, workflow subcommands, output renderer, exit mapper, progress reporter, cancellation handling, and GUI-CLI parity tests.
11. Add audit views, complete deterministic tests, define KUMA requirements, analyze failures, improve the system, and retest.

MVP fallback. The mandatory end-to-end slice retains both the browser GUI and supported CLI. It includes program scope, hardened Burp XML import, evidence storage, attack-surface browsing, local methodology import, the generic vulnerability-module contract, the XSS reference module, bounded planning, validation, and coverage. HAR import, GraphQL-specific enrichment, account comparison, direct Notion synchronization, simultaneous GUI-CLI operation, and additional vulnerability modules remain priority-two work.

13.3 Planned Extensions

NotionKnowledgeSource will synchronize changed pages through the same KnowledgeSource port. A Montoya-based Burp extension will later contribute selected requests through TrafficImporter or a dedicated ingestion port. A later CLI client mode may connect to the running loopback server for simultaneous use. New access-control and IDOR, SQL injection, SSRF, authentication and session, CSRF, file-upload, GraphQL, business-logic, and race-condition modules will implement VulnerabilityModule and reuse shared traffic, knowledge, scope, planning, validation, provenance, and coverage infrastructure.

14 Stage 1 Completion Checklist

Assignment artifact	Location in this report	Status
Project description and scope	Sections 1 and 3	Complete
Major features	Section 2.3	Complete
Use cases	Section 5.2	Ten complete specifications including CLI operation
UML use case diagram	Figure 5	Complete
Sequence diagrams	Figures 8 through 13	Six major flows with alternatives
UML class diagram	Figures 14 through 19	Six readable views of one logical design
GUI and CLI requirements	Sections 2, 4, 6, 8, 10, and 11	Architecture and traceability complete
Feature to class mapping	Section 10	Complete
Important design decisions	Section 9.1	Complete
Software design principles	Section 9	Complete
Traceability foundation	Section 11	Requirements to use cases, sequences, and classes

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Glossary

Term	Meaning in this design
Attack surface	The observed hosts, endpoints, methods, parameters, content types, workflows, and account contexts available for investigation.
EvidenceRef	A stable identifier that points from a normalized observation or recommendation to the original captured record.
KnowledgeCitation	A reference to a specific version and section of an imported methodology source.
Methodology memory	The versioned, indexed body of the hunter's notes and approved technique records.
TestIdea	An evidence-grounded hypothesis and manual investigation plan, not a confirmed vulnerability.
Coverage	The recorded state of testing for an endpoint, vulnerability category, and account context.
Agent tool	A typed and validated read-only operation through which the LLM queries project data.
Scope decision	The deterministic result in scope, out of scope, or unresolved for an asset or record.
Vulnerability module	A category-specific deterministic analyzer that converts immutable scoped evidence into ModuleAnalysis through a shared interface.
ModuleAnalysis	A category-neutral result containing applicability, observed signals, EvidenceRefs, prerequisites, and a knowledge query; it is not a finding.
Reference module	The first complete implementation used to validate the shared contract. XSS is the course-release reference module, not the platform boundary.
CLI	The supported Picocli command-line interface through which the hunter invokes major HuntAssist workflows interactively or from scripts.
Inbound adapter	A GUI controller or CLI command that translates user input into shared application requests and renders sanitized application results.
Exit-code category	A documented stable CLI result category such as success, invalid arguments, policy refusal, dependency unavailable, store busy, or cancellation.
Interface parity	The requirement that equivalent GUI and CLI operations use the same application services and enforce the same safety and persistence rules.